

Addressing Information Asymmetry: Deep Temporal Causality Discovery for Mixed Time Series

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Abstract—While existing causal discovery methods mostly focus on continuous time series, causal discovery for mixed time series encompassing both continuous variables (CVs) and discrete variables (DVs) is a fundamental yet underexplored problem. Together with nonlinearity and high dimensionality, mixed time series pose significant challenges for causal discovery. This study addresses the aforementioned challenges based on the following recognitions: 1) DVs may originate from latent continuous variables (LCVs) and undergo discretization processes due to measurement limitations, storage requirements, and other reasons. 2) LCVs contain fine-grained information and interact with CVs. By leveraging these interactions, the intrinsic continuity of DVs can be recovered. Thereupon, we propose a generic deep mixed time series temporal causal discovery framework. Our key idea is to adaptively recover LCVs from DVs with the guidance of CVs and perform causal discovery in a unified continuous-valued space. Technically, a new contextual adaptive Gaussian kernel embedding technique is developed for latent continuity recovery by adaptively aggregating temporal contextual information of DVs. Accordingly, two interdependent model training stages are devised for learning the latent continuity recovery with self-supervision and causal structure learning with sparsity-induced optimization. Experimentally, extensive empirical evaluations and in-depth investigations validate the superior performance of our framework.

Index Terms—Time series causal discovery, Granger causality, information asymmetry, mixed time series, contextual adaptive Gaussian kernel embedding, self-supervised learning.

I. INTRODUCTION

GOING beyond classification and prediction, recognizing causal structures from time series has become an increasingly critical research topic in recent decades [1], [2], [3]. By identifying and interpreting the underlying interactions of the observational data, causal discovery is energized in various real-world applications, such as ecosystems [4], neuroscience [5],

climatology [6] and visual understanding [7] due to its practicality, efficiency, and interpretability.

Built upon an intuitive recognition that “*past causing the future*”, Granger causality (GC) [8] has been developed as a pioneering and prevalent causal discovery paradigm. GC leverages the temporal ordering inherent to time series to draw causal existence and directions, which has been proven to be effective and spawned plentiful remarkable works [9], [10], [11]. Especially in the recent deep learning community, many advanced methods, such as Neural Granger Causality [12], eSRU [13], GVAR [14] and CR-VAE [15] have been proposed to cope with multivariate nonlinearity by utilizing the powerful expressive capacity of neural networks and effective optimization techniques.

Despite the remarkable progress, most causal discovery methods are developed with an inherent assumption that the time series data are solely composed of continuous-valued variables. However, in many practical scenarios, the collected data are often a mixture that encompasses discrete variables (DVs) and continuous variables (CVs), which are referred to as mixed data [16]. In industrial systems, some process variables, e.g., temperature and flow rate, are often measured with continuous values, while some associated alarm signals are recorded in discrete forms. In microservice systems, both continuous KPI indicators (e.g., CPU and memory usage) and discrete event sequences are stored for stability maintenance. So far, causal analysis for mixed data is still an emerging yet formidable task. Specifically, directly identifying the causal relationships between the DVs and CVs is difficult due to the discrepancy of distribution types and the information granularity gap between DVs and CVs. In this paper, we ascribe the above key challenges as *Information Asymmetry* (InfoA) problem. When encountering mixed data, a straightforward solution is to ignore the DVs and apply the methods only for CVs to obtain a partial causal graph. Alternatively, one may design discretization policies to discretize the CVs to infer discrete causal networks [17], [18]. However, these methods may lose vital information or sacrifice fine-grained continuous signals, thereby inevitably yielding non-negligible estimation errors. Besides, several studies [16], [19], [20] have made attempts for causal discovery among DVs and CVs with specific applications via constraint-based or score-based algorithms. To date, these methods are mostly designed for non-temporal tabular data and may be restricted to linear or bivariate cases in specific domains. However, real-world MiTS often emerge with intricate nonlinearity and high dimensionality, which further exacerbates the difficulties for accurate causal discovery.

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Our code and data are available at <https://github.com/chunhui/MiTCD>.

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Essentially, due to measurement difficulties, storage efficiency, and other limitations or purposes, many intrinsically continuous-valued signals are often recorded with discrete-valued forms, e.g., meteorological data, product sales volume, stock returns, and health states [21]. Given an example in industrial sensing systems, some hard-to-measure pressure variables are recorded as binary alarm signals based on whether they exceed the control limits [22]. Inspired by the above circumstances, we recognize that the DVs may originate from latent continuous variables (LCVs). Accordingly, previous studies [16], [23] follow the settings that the observed DVs are derived from underlying LCVs via discretization. The intrinsic continuous-valued information of LCVs is unobservable. Also, there exist information granularity gaps between DVs and CVs, as the discretization causes an information loss of LCVs [24] and changes the distribution type. These discrepancies exacerbate the challenge of directly identifying causal relationships between DVs and CVs. However, in the dynamic systems, the LCVs of corresponding DVs often have interactions with the observed CVs, e.g., the pressure alarm signals are potentially triggered by associated temperature indicators and change the related flow rates within the control loop. These interactions can be leveraged to recover the corresponding LCVs from DVs with pertinently designed approaches. During the recovery process, the information granularity gap between DVs and CVs can be gradually bridged and the distribution types can be unified. In this way, causal discovery can be performed in a unified continuous-valued space.

Enlightened by the aforementioned analysis, we present MiTCD, a deep **M**ixed Time Series **C**ausal **D**iscovery framework. Different from existing methods that neglect DVs or discretize CVs, MiTCD addresses the InfoA problem from a new but intuitive perspective, i.e., recovering the latent continuity of DVs by leveraging the interactions between CVs and LCVs. Besides, empowered by sparse-induced neural networks, MiTCD can identify causal structures under nonlinear and high-dimensional scenarios. Concretely, we design a Contextual Adaptive Gaussian Embedding (CAGKE) technique to portray and recover the latent continuity of DVs by aggregating multi-scale contextual temporal information of DVs with weighted incentive Gaussian kernels (As illustrated in the *Right* of Fig. 1). The parameters of CAGKE are adaptatively learned in a self-supervised manner via the forecasting task of CVs. On the basis of pre-trained CAGKE, we conduct causal discovery among LCVs and CVs in a unified continuous-valued space. Technically, sparse constraints, i.e., hierarchical group lasso penalties are imposed on the forecasting networks to select significant causal relationships. Furthermore, Proximal Gradient Descent (PGD) algorithm is leveraged to optimize the sparsified-induced loss term for obtaining interpretable and explicit sparse causal structures of MiTS. Our contributions are summarized in three folds:

- 1) We refine that the InfoA problem is essential in MiTS causal discovery and address it from a new perspective, i.e., adaptively recovering the LCVs from DVs with the guidance of CVs to bridge the information granularity gap and unify the distribution types. In this way, causal

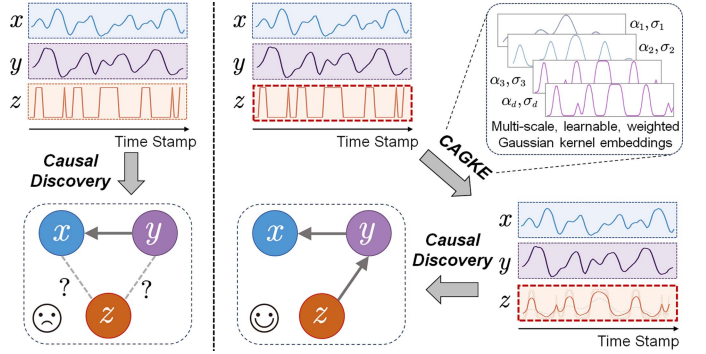


Fig. 1. Contrast comparison of MiTCD and general causal discovery approaches. Kindly note: x and y are CVs, and z is a DV. *Left*: illustration of general approaches that struggle to concurrently analyze mixed time series due to the discrepancy of distribution types; *Right*: MiTCD that adopts *multi-scale, learnable, and weighted incentive Gaussian kernels* to adaptively recover the latent continuity of DVs and perform causal discovery in a unified continuous-valued space.

discovery can be performed in a unified continuous-valued space for CVs and LCVs.

- 2) A new CAGKE technique is developed by introducing learnable and weighted incentive Gaussian kernels, which can aggregate multi-scale temporal contextual information to re-establish continuity signals for DVs and alleviate the negative effects of the InfoA problem.
- 3) Built upon CAGKE, corresponding training strategies are pertinently designed. On the one hand, the parameters of CAGKE can be adaptively learned in a self-supervised manner via the guidance of CVs. On the other hand, sparse and significant causal relations can be revealed via sparsity-induced constraints and optimization techniques.

II. RELATED WORKS

In this section, we mainly focus on the existing techniques for Granger causality discovery and mixed data modeling and causal discovery approaches.

A. Granger Causality Discovery

Granger Causality (GC) is a pioneering and predominant causal discovery paradigm that leverages predictive contributions as matrices for relation inference of time series [8], [25]. Primordially, original GC is developed but restricted to pairwise and linear cases and relies on statistical analysis. Later on, numerous advances have pushed the envelope on where GC can be applied by loosening these restrictions in a variety of ways. Specifically, conditional Granger [26] and Lasso Granger Causality (LGC) [9] extend the GC to multivariate scenarios based on autoregressive models and various sparse regularizers. More recently, researchers have leveraged the expressive power of deep neural networks for nonlinear GC discovery [10], [27]. Neural Granger Causality [12] is proposed with sparse-input neural networks to model nonlinear autoregressive dynamics within time series. Literature [13] presents a new economy statistical recurrent unit (eSRU) to infer the topology of GC

relations. By leveraging attention-based CNN, temporal causal discovery framework (TCDF) [28] devises systematic criteria for GC determination. Empowered by recurrent VAE and sparse penalties, CR-VAE [15] is designed to learn a sparse causal matrix to encode the GC relations for data generation. Aided by graph convolutional network (GCN), NTiCD [29] and MPGE [30] adopt learnable adjacency matrices to learn non-linear and multivariate GC relationships.

Apart from GC, score-based methods have also been developed for temporal causal discovery. VARLiNGAM [31] combines the non-Gaussian instantaneous model with autoregressive models to estimate structural vector autoregressive models. Later on, MS-CASTLE [32] is proposed to estimate the structure of linear causal relationships occurring at different time scales, with a simplified single-scale version named SS-CASTLE. Dynotears [33] estimates instantaneous and time-lagged relationships between time series via a smooth equality constraint. Following, NTSnotears [34] extends Dynotears to nonlinear scenarios by utilizing 1-D CNNs. Besides, mutual information (MI) and conditional independence (CI) tests can also be leveraged as criteria to determine temporal causal relations with nonlinearity, typical methods include PCMC [6], [35] and PCTMI [36].

However, most of the above methods are tailored for continuous time series and are therefore inappropriate for MiTS. In this paper, we propose a systematic framework for inferring GC relationships for MiTS through concurrent analysis of DVs and CVs.

B. Mixed Data Modeling and Causal Discovery

Mixed data widely exist in various application domains, such as medical diagnosis [19], process industry [37], server maintenance [38], and economy pricing [20]. Mixed data modeling and analysis are critical yet challenging research tasks. Particularly for causal discovery, PCTLC [39] is developed for evaluating the significance of causes of continuous-valued effects using conditional expectations for stock price analysis. Later on, [19] proposed a score-based algorithm termed CausalMGM to identify causal variables for disease diagnosis. Towards prescriptive pricing, [20] developed a mixed causal model termed MIC and proved its identifiability in the bivariate cases. Beyond this work, Linear Mixed causal model (LiM) [16] is proposed, which extends sufficient identifiability conditions to multivariate cases. Besides, Hector-PC [40] is designed as a conditional independent (CI) test approach for mixed data by pair-wise correlation estimation, which tests the rank correlation between two CVs, the polyserial correlation between a CV and a DV, and the polychoric correlation between two DVs. By leveraging appropriate regression model-based likelihood-ratio tests and CI tests, RegressionCI (RegCI) [41] is designed to discover causal relations from mixed data. Recently, Runge et.al. [1] developed a comprehensive library Tigramite for temporal causal discovery, in which RegCI acts as a CI indicator and can be combined with advanced causal discovery algorithms.

To sum up, existing causal inference methods for mixed data are typically developed using constraint-based or score-based

algorithms, which are primarily suited for tabular data and may be limited to linear or bivariate cases in certain domains. Fortunately, our proposed MiTCD approach can handle multivariate nonlinear temporal causal relationships by utilizing the designed CAGKE technique and sparse-induced neural networks under the GC paradigm.

III. METHODOLOGY

In this section, we first present the problem formulation of MiTS causal discovery. Then, the motivation of this study is analyzed with relevant discussions. Following, we elaborate on the proposed MiTCD framework from its overview to its implementation details.

A. Problem Setup and Motivation Analysis

1) *Problem Formulation and Notations*: Consider a nonlinear dynamic system that is characterized by p observed univariate time series with length T . We denote the whole sequences as $\mathbf{x} = \{x_1, x_2, \dots, x_p\} \in \mathbb{R}^{p \times T}$. Among them, there are $p - n$ continuous variables $\mathbf{x}^C = \{x_1, x_2, \dots, x_{p-n}\} \in \mathbb{R}^{(p-n) \times T}$. Without loss of generality, the value of continuous variables is normalized between 0 to 1. Besides, there are n discrete variables $\mathbf{x}^D = \{x_1, x_2, \dots, x_n\} \in \mathbb{A}^{n \times T}$. Following the previous works of [16], [23], we consider the discrete variables that originate from latent continuous variables through discretization processes. We denote the number of discrete states as N_{DV_s} . The value of the discrete variables ranges within the integers between 0 and $N_{DV_s} - 1$, i.e., $\mathbb{A} = \{0, 1, \dots, N_{DV_s} - 1\}$.

The goal of MiTS causal discovery is to obtain a causal summary graph from the observed time series data. The causal graph can be represented by a directed adjacency matrix A , in which $A_{(i,j)}$ denotes whether the variable x_j significantly causes the variable x_i (denoted by 1) or not (denoted by 0). We follow the definition of generalized Granger Causality [10], [12] as the criterion for causal relationship determination, which is presented as follows.

Definition 1 (Granger Causality): Given a predictive function $x_{t,j} = g_j(x_{<t,1}, x_{<t,2}, \dots, x_{<t,p}) + e_{t,j}$, where $x_{<t,i} = \{x_{t-k,i}, \dots, x_{t-2,i}, x_{t-1,i}\}$ represents the lagged information of variable x_i with lag k , and e_t represents the residuals. Formally, variable x_i significantly causes x_j if g_j is not invariant to $x_{<t,i}$. Mathematically,

$$\begin{aligned} \text{if } \exists x'_{<t,i} \neq x_{<t,i} : g_j(x_{<t,1}, \dots, x'_{<t,i}, \dots, x_{<t,p}) \\ \neq g_j(x_{<t,1}, \dots, x_{<t,i}, \dots, x_{<t,p}) \end{aligned} \quad (1)$$

In other words, variable x_i Granger cause to x_j means that the predictive function g_j is dependent on x_i . Following, we discuss the assumptions under which Granger causality holds. The definitions of the necessary assumptions are given as follows.

Definition 2 (Causal Sufficiency): A set of variables is said to be causally sufficient if all common causes of all variables are observed.

Definition 3 (Temporal Priority): A causal relation between two variables is said to satisfy the temporal priority if it is oriented in such a way that the cause occurred before its effect.

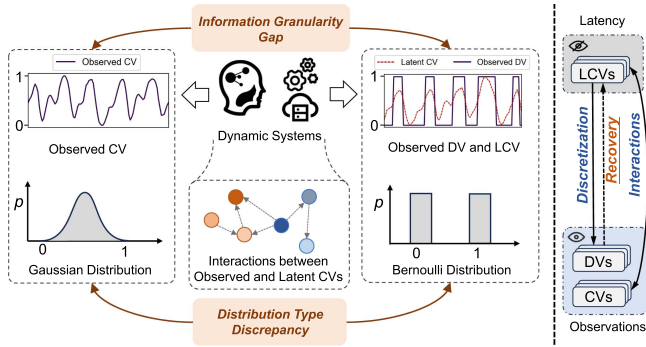


Fig. 2. Illustration of MiTS and InfoA problem. *Left*: Dynamic systems involve multiple variables that interact with each other. Some variables can be observed with their intrinsic continuous values and are known as CVs, while others are unobservable and are termed LCVs, which are discretized for storage and processing and can be observed as DVs; *Right*: Connections between CVs, DVs, and LCVs.

Specifically, causal sufficiency [42] (Definition 2) requires the observed variables to be a complete representation of the system considered, i.e., there are no unmeasured confounders, while temporal priority [43] (Definition 3) makes the process of causality asymmetric in time. In addition, we assume no contemporaneous causal effects for the present time series. More discussions of the assumptions of Granger causality can be found in existing papers focusing on time series causal discovery [10], [42].

2) *Motivation Analysis and Discussions*: Previous studies have highlighted the urgency and significance of causal discovery for MiTS and made several attempts from various perspectives. However, as discussed earlier, there still exist unsolved deficiencies that widely occur in real-world applications. In this paper, we ascribe the unsolved challenges to the following two aspects.

Statement 1: For the mixed time series, the discrepancy of distribution types and the gap of information granularity between discrete variables and continuous variables lead to the Information Asymmetry problem, which impedes directly discovering causal relations among the discrete variables and continuous variables.

As presented in Fig. 2, the InfoA problem is caused by two interdependent factors. First, among MiTS, the DVs often follow Bernoulli distributions while the CVs approximately follow Gaussian distribution, which yields the discrepancy of distribution types. Here we define “variable heterogeneity” as the presence of different kinds of variables (i.e., CVs and DVs) that follow distinct distribution types. Second, due to multiple factors, DVs often originate from LCVs through discretization processes, during which vital continuity information is lost, thus leading to the information granularity gap between DVs and CVs. However, primary causal discovery methods typically assume that all observed variables follow a common distribution. Besides, they determine causal relationships by evaluating and quantifying information contributions between variables [1], [10]. Inevitably, InfoA would make it difficult to directly identify causal relationships between DVs and CVs, and yield non-negligible estimation errors.

TABLE I
COMPARISON OF THE PROPOSED MiTCD AND THE STATE-OF-THE-ART CAUSAL DISCOVERY METHODS

	Methods	Temporal	Nonlinear	Multivariate	Heterogeneous
Continuous data	LGC [9]		×	✓	×
	VARLiNGAM [31]	✓	×	✓	×
	PCMCi [6], [35]	✓	✓	✓	×
	cMLP, cLSTM [12]	✓	✓	✓	×
	eSRU [13]	✓	✓	✓	×
	GVAR [14]	✓	✓	✓	×
	PCTMI [36]	✓	✓	✓	×
	Dynotears [33]	✓	×	✓	×
	NTSnotears [34]	✓	✓	✓	×
	SS-CASTLE [32]	✓	×	✓	×
	MS-CASTLE [32]	✓	×	✓	×
	CR-VAE [15]	✓	✓	✓	×
Mixed data	NTICD [29]	✓	✓	✓	×
	PCTLC [39]	✓	×	×	✓
	MIC [20]	×	×	×	✓
	LiM [16]	×	×	✓	✓
	Hetcor-PC[40]	×	✓	×	✓
	RegCI [41][1]	✓	×	✓	✓
	MiTCD (Ours)	✓	✓	✓	✓

Kindly note: “Temporal” denotes whether the method is suitable for time series. “Multivariate” represents whether the method is designed for multivariate scenarios or needs pairwise analysis. “Heterogeneous” denotes whether the method is specifically designed to cope with variable heterogeneity, i.e., can handle different variables with different distribution types.

Statement 2: Together with variable heterogeneity, real-world mixed time series often encompass a large number of variables that often exhibit nonlinear inter-variable causal relationships, which further exacerbates the difficulties for mixed time series causal discovery and puts forward a formidable research gap.

In practice, many interactions such as those in neuroscience and industry systems are inherently nonlinear and involve multiple variables [12], [27], which poses significant challenges for MiTS causal discovery. Particularly, Table I reviews current state-of-the-art causal discovery methods and compares them with our proposed MiTCD. On the one hand, current deep learning-based approaches can well handle nonlinear and multivariate scenarios by leveraging powerful neural networks. However, they are inappropriate for MiTS because of the InfoA problem. On the other hand, current MiTS causal discovery approaches model the heterogeneous distributions of DVs and CVs based on certain assumptions, which may limit their feasibility in characterizing complex nonlinearity and may suffer from high-dimensional settings. Therefore, *our approach aims at filling the research gap to concurrently identify multivariate and nonlinear temporal causal relationships of MiTS under the GC paradigm in a coherent framework.*

B. Overview of the Proposed MiTCD

The overall framework of MiTCD is presented in Fig. 3, which is composed of Contextual Adaptive Gaussian Kernel Embedding (CAGKE) technique and two different but interdependent training stages.

CAGKE: The core idea of CAGKE is to recover the latent continuity of DVs to characterize the constitutional temporal dynamics. By introducing multi-scale learnable incentive Gaussian kernels with diverse bandwidth and weighted

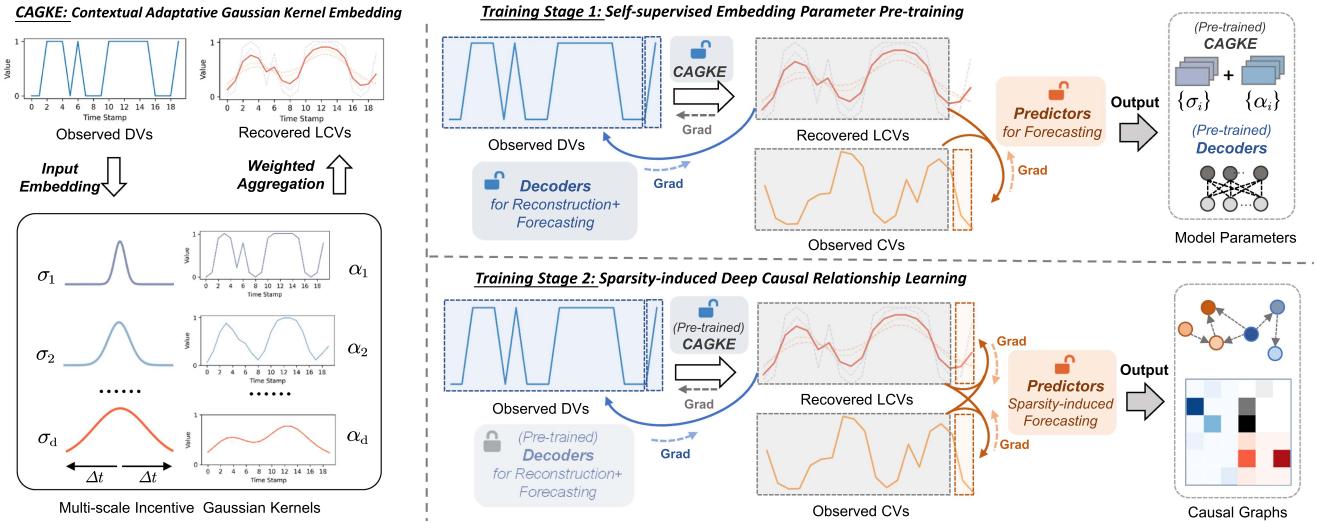


Fig. 3. The overall framework of MiTCD, including (1) CAGKE for recovering LCVs from DVs, (2) self-supervised embedding parameter pre-training stage for CAGKE parameter learning under the guidance of CVs, and (3) sparsity-inducing deep causal relationship learning stage for causal structure identification of MiTS. Kindly note: “Grad” denotes the backpropagating gradients. The “unlock” icon means the corresponding module is trainable while the “lock” icon means the corresponding module is untrainable with fixed parameters.

aggregation approaches, temporal contextual information can be re-incorporated to enrich the information granularity and transform the distribution type of DVs, thereby alleviating the InfoA problem and facilitating causal discovery.

Self-supervised Pre-training Stage: To obtain the optimal CAGKE parameters, the first training stage is devised in a self-supervised manner, in which models explore supervision signals from the future states of CVs through forecasting tasks. Guided by the objective to minimize the forecasting error, CVs can provide useful information for CAGKE parameter learning thereby empowering latent continuity recovery from DVs. Meanwhile, discrete variable decoders are designed to reconstruct the observed DVs using the obtained LCVs to ensure that the recovery process is reversible.

Sparsity-induced Causal Learning Stage: Built upon the well-trained CAGKE and discrete variable decoders, the second training stage is devised with sparsity-inducing regularization and corresponding optimization techniques for causal relationship learning. In this training stage, the CVs and LCVs obtained by CAGKE are concurrently fed into the sparse input predictor networks under the constraints of hierarchical group-lasso penalties. By leveraging the proximal gradient descent (PGD) algorithm, significant causal relationships can be selected to compose overall causal summary graphs for MiTS.

In the following, all the details of model components and training strategies will be elaborately introduced.

C. Contextual Adaptive Gaussian Kernel Embedding

As mentioned above, DVs may originate from LCVs and undergo the discretization process, during which the information granularity has been compromised and the distribution type has been changed. To address the InfoA problem, CAGKE is designed to recover the intrinsic continuity information for DVs. Inspired by density estimation [44], we adopt the Gaussian

kernels to re-establish the continuous temporal dynamic signals of DVs because:

1) Gaussian kernels have smoothness properties to characterize temporal adjacent-concentration constitutionally [45], as the kernel $\mathcal{K}(x, x') = \exp(-\|x - x'\|^2 / 2\sigma^2)$ smoothly decays with the increasing temporal distance $\|x - x'\|^2 = \text{var} \Delta t$, ensuring that nearby temporal points exhibit similar values. In addition, it can fit the function with complex shapes and distributions [46], [47];

2) By controlling the scale of bandwidth, it can aggregate multi-scale temporal contextual information, e.g., small bandwidth can influence local short-term range time stamps while large bandwidth can involve long-term temporal impacts. (as illustrated in the bottom left of Fig. 3)

Contextual Incentive Gaussian Kernels Formally, given a DV x_j^D , every time point of non-zero value can be regarded as an incentive signal. At each incentive point, a Gaussian kernel with bandwidth σ is superimposed around this time stamp as

$$f(\Delta t; \sigma) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{\Delta t^2}{2\sigma^2}\right) \quad (2)$$

where σ is the standard deviation, which can also be interpreted as the bandwidth of the Gaussian kernel; Δt^2 denotes the distance from the incentive signal.

Taking all incentive time points into consideration, a continuous-valued time series can be obtained with the superposition of all incentive Gaussian kernels as

$$F_{x_j^D}(t; \sigma) = \sum_{t_i=1}^T x_{t_i,j}^D \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(t - t_i)^2}{2\sigma^2}\right) \quad (3)$$

Multi-scale Adaptive Weighted Aggregation Inspired by the observation that multi-scale temporal information is a natural and critical property of time series, multiple-sized Gaussian incentive kernels are designed to integrate multi-scale temporal

information of the DVs. Moreover, the multiple-sized Gaussian incentive kernels are aggregated with corresponding weights to characterize their varying impacts. Specifically, the value of the recovered LCVs can be calculated as follows

$$x_{t,j}^{LC} = \sum_{i=1}^d \hat{\alpha}_i F_{x_j^D}(t; \sigma_i) + \varepsilon_t \quad (4)$$

where $\{\sigma_1, \sigma_2, \dots, \sigma_j, \dots, \sigma_d\}$ are bandwidth parameters to characterize multi-scale temporal patterns and d is the number of multi-scale incentive Gaussian kernels. ε_t is the noise term that follows normal distribution as $\varepsilon_t \sim \mathcal{N}(0, 0.01)$. Without loss of generality, the bandwidth parameters are arranged from small to large as $\{\sigma_{\min} = \sigma_1 < \sigma_2 < \sigma_j < \dots < \sigma_d = \sigma_{\max}\}$. Specifically, $\sigma_{\min}$ and $\sigma_{\max}$ are learnable parameters that control the upper and lower bound of bandwidth, respectively. The other bandwidth parameters are generated as an arithmetic sequence as

$$\{\sigma_i\} = \sigma_{\min} + (i-1) \frac{\sigma_{\max} - \sigma_{\min}}{d} \quad (5)$$

Correspondingly, $\{\hat{\alpha}_1, \hat{\alpha}_2, \dots, \hat{\alpha}_j, \dots, \hat{\alpha}_d\}$ are learnable weight coefficients that correspond to every incentive Gaussian kernels, which are normalized by Softmax operation as follows:

$$(\hat{\alpha}_1, \hat{\alpha}_2, \dots, \hat{\alpha}_d) = \text{Softmax}(\alpha_1, \alpha_2, \dots, \alpha_d) \quad (6)$$

In the training procedure, the weight coefficients α are all initialized with 1 and can be learned adaptively to characterize different importance of multi-scale incentive Gaussian kernels.

D. Self-Supervised Pre-Training of Embedding Parameters

Through CAGKE, DVs can be recovered into LCVs as (7) for further causal discovery. In the following, we will introduce the training procedure for CAGKE parameter learning via self-supervision.

$$\tilde{x}_j^{LC} = \text{CAGKE}(x_j^D; \{\sigma_1, \dots, \sigma_d; \alpha_1, \dots, \alpha_d\}) \quad (7)$$

Technically, the design of training strategies is based on two following recognitions and principles:

1) *By leveraging the causal relationships between CVs and LCVs, CVs can provide instructive information for recovering the latent continuity from observed DVs, which are further discussed in Theorem 1 in Section IV-B.*

2) *The recovered LCVs should retain information about the observed DVs i.e., $\tilde{x}_j^{LC}$ can well reconstruct the original x_j^D through a function as $x_j^D = g(\tilde{x}_j^{LC})$.*

Motivated by the above analysis, two data forward branches and corresponding optimization objectives are designed.

Data Feedforward Branches In the main forward branch, the past values of recovered LCVs and the past values of the CVs are fed into predictors (denoted as Predictor) to forecast the future states of the CVs. In this way, the future values of CVs can act as supervision signals to guide the learning of CAGKE parameters. In the other branch, the past values of recovered LCVs are fed into discrete variable decoders (denoted as Decoder) to reconstruct the original DVs. The reconstruction loss can guarantee that the transformation of CAGKE would not

pay overabundant attention to the forecasting of CVs and can still preserve original discrete-valued information. Here every loss term is introduced as follows:

Forecasting Loss Denote the forecasting results of the CVs as $\hat{x}_{j,t}^C$, the forecasting error loss can be formed as

$$\mathcal{L}_{\text{fore}} = \sum_{j=1}^{p-n} \text{MSE}(\hat{x}_{j,t}^C, x_{j,t}^C) \quad (8)$$

where $\hat{x}_{j,t}^C = \text{Predictor}_j(\{x_{i,t-\tau:t-1}^C\}_{i=1}^{p-n}; \{\tilde{x}_{i,t-\tau:t-1}^{LC}\}_{i=1}^n)$, and MSE represents mean square error loss function.

Reconstruction Loss Denote the reconstruction results of the DVs as $\hat{x}_{j,t-k:t}^D$, the reconstruction loss can be formulated as

$$\mathcal{L}_{\text{recons}} = \sum_{j=1}^n \text{CE}(\hat{x}_{j,t}^D, x_{j,t}^D) \quad (9)$$

where $\hat{x}_{j,t}^D \leftarrow \text{Decoder}_j(\tilde{x}_{j,t-\tau:t}^{LC})$, and CE represents cross-entropy loss.

Finally, the overall loss term of the first training stage can be formulated as

$$\mathcal{L}_{\text{all}} = \lambda_1 \mathcal{L}_{\text{fore}} + \lambda_2 \mathcal{L}_{\text{recons}} \quad (10)$$

where λ_1 and λ_2 are weighted coefficients.

The overall procedure of the self-supervised pre-training stage is depicted in Algorithm 1. After the training procedure, we can obtain the well-optimized parameters of CAGKE and discrete variable decoders, which serve as bridges between the two training stages.

E. Sparsity-Inducing Deep Causal Relation Learning

Built upon the first training stage, the second training stage aims at learning sparse causal structures from MiTS via sparsity-inducing optimization.

Pre-trained Parameter Transfer In the first training stage, parameters of CAGKE are well-trained via the supervision of CVs, and the discrete variable decoders are trained to reconstruct the observed DVs well. Therefore, the parameters of pre-trained CAGKE and discrete variable decoders are used as the initialization in the second training stage. Specifically, we fix the parameters of discrete variable decoders and bandwidth σ_i of CAGKE to ensure training stability and we keep the weight parameters α_i trainable to adaptively facilitate causal discovery. Then, both the recovered LCVs $\tilde{x}^{LC}$ and original CVs x^C are equally and concurrently fed into individual predictors to accomplish the forecasting tasks under the paradigm of GC. The optimization objective contains the following parts:

Forecasting Error Loss The error loss term is designed to minimize the forecasting loss as

$$\mathcal{L}_{\text{fore}} = \sum_{j=1}^{p-n} \text{MSE}(\hat{x}_{j,t}^C, x_{j,t}^C) + \sum_{j=1}^n \text{MSE}(\hat{x}_{j,t}^{LC}, \tilde{x}_{j,t}^{LC})$$

where $\hat{x}_{j,t}^C$ and $\hat{x}_{j,t}^{LC}$ denote the output prediction value.

Discrete Variable Reconstruction Loss Since the weight parameter α_i is trainable, the discrete variable reconstruction loss

Algorithm 1: Pre-Training for CAGKE Parameter Learning:

Input: $\mathbf{x}^C = \{x_1, x_2, \dots, x_{p-n}\} \in \mathbb{R}^{(p-n) \times T}$,
 $\mathbf{x}^D = \{x_1, x_2, \dots, x_n\} \in \mathbb{R}^{n \times T}$, τ , λ_1 and λ_2 .
Output: Pre-trained model parameters of
CAGKE_i ($\sigma_{\min}, \sigma_{\max}; \alpha_1, \dots, \alpha_d$) and
Decoder_i ($i = 1, 2, \dots, n$).
1: Initialize the parameters of the models;
2: **while** not converge **do**
3: // Model Feedforward
4: **for** i in range(n) **do**
5: Recovering the LCVs for DVs
 $\tilde{x}_i^{LC} \leftarrow \text{CAGKE}_i(x_i^D)$;
6: Reconstruction of DVs $\hat{x}_{i,t}^D \leftarrow \text{Decoder}_i(\tilde{x}_{i,t-\tau:t}^{LC})$
7: **end for**
8: **for** i in range ($p - n$) **do**
9: Predict the future value of CVs $\hat{x}_{i,t}^C \leftarrow$
 $\text{Predictor}_i(\{x_{j,t-\tau:t-1}^C\}_{j=1}^{p-n}; \{\tilde{x}_{j,t-\tau:t-1}^{LC}\}_{j=1}^n)$
10: **end for**
11: // Loss Calculation
12: Forecasting error loss $\mathcal{L}_{\text{fore}} \leftarrow \sum_{j=1}^{p-n} \text{MSE}(\hat{x}_{j,t}^C, x_{j,t}^C)$.
13: Reconstruction loss $\mathcal{L}_{\text{recons}} \leftarrow \sum_{j=1}^n \text{CE}(\hat{x}_{j,t}^D, x_{j,t}^D)$.
14: Overall loss $\mathcal{L}_{\text{all}} \leftarrow \lambda_1 \mathcal{L}_{\text{fore}} + \lambda_2 \mathcal{L}_{\text{recons}}$
15: // Parameter Update
16: Update the parameters of models using Adam
optimizer to minimize $\mathcal{L}_{\text{all}}$;
17: **end while**
18: **return** Pre-trained model parameters.

is also deployed to guarantee the transformation to be reversible as

$$\mathcal{L}_{\text{recons}} = \sum_{j=1}^n \text{CE}(\hat{x}_{j,t}^D, x_{j,t}^D) \quad (11)$$

Sparsity-inducing Constraints To obtain sparse and significant causal structures, sparse-inducing constraint, i.e., hierarchical group lasso, is applied to the first layer of each predictor network as

$$\mathcal{L}_{\text{spar}} = \sum_{j=1}^p \sum_{t=1}^k \|W_{:j}^t, \dots, W_{:j}^k\|_2 \quad (12)$$

where W is the coefficient matrix of the first layer that corresponds to each predictor of the variable x_i to be predicted.

Finally, the overall loss term of the second training stage can be formulated as

$$\mathcal{L}_{\text{all}} \leftarrow \lambda_1 \mathcal{L}_{\text{fore}} + \lambda_2 \mathcal{L}_{\text{recons}} + \lambda_3 \mathcal{L}_{\text{spar}} \quad (13)$$

where λ_1 , λ_2 and λ_3 are weighted coefficients.

Proximal Gradient Descent-aided Optimization Moreover, to ensure the sparsity of the optimization results, Proximal Gradient Descent (PGD) algorithm [48], [49] is adopted to optimize the training objective. It is because PGD can shrink relatively small parameters to zero regularly through its soft threshold mechanism, thus ensuring the sparsity of parameters, which is an ideal reflection of the Granger non-causality between variables.

Concretely, denote the trainable parameters as θ , PGD algorithm updates the parameters iteratively starting with $\theta^{(0)}$ by

$$\theta^{(m+1)} = \text{prox}_{l_{r(m)} \lambda_3 \Omega} \left(\theta^{(m)} - l_{r(m)} \nabla \mathcal{L}_{\text{smo.}} \left(\theta^{(m)} \right) \right) \quad (14)$$

where $\text{prox}_{\lambda_3 \Omega}$ is the proximal operator with respect to the sparsity-induced penalty function; $\mathcal{L}_{\text{smo.}} = \lambda_1 \mathcal{L}_{\text{fore}} + \lambda_2 \mathcal{L}_{\text{recons}}$ is the smooth loss term; m is the index of training epoch. ∇ denotes the calculation of the gradient.

Specifically for the penalized parameters, the proximal step is given by a group soft-thresholding operation that applied to the first layer of each predictor network as

$$\begin{aligned} & \left\{ \left\{ (W_{:j}^t, \dots, W_{:j}^{1k})^{(m+1)} \right\}_{j=1}^{j=p} \right\}_{t=1}^{t=k} \\ &= \text{soft} \left((W_{:j}^t, \dots, W_{:j}^{1k})^{(m*)}, \lambda_3 l_{r(m)} \right) \\ &= \left(1 - \frac{\lambda_3 l_{r(m)}}{\left\| (W_{:j}^t, \dots, W_{:j}^{1k})^{(m*)} \right\|_2} \right) (W_{:j}^t, \dots, W_{:j}^{1k})^{(m*)} \end{aligned} \quad (15)$$

where $(x)_+ = \max(0, x)$.

The overall procedure of the deep causal learning stage is depicted in Algorithm 2. Besides, the detailed implementation of PGD can be found in Appendix, available online. Finally, the integral causal structure can be interpreted by checking the parameter magnitude of each predictor.

Causal Summary Graph Interpretation For a variable x_i to be predicted, the element value of $W_{:j}^t$ reflects whether the variable x_j at lag t can help predict the future value of x_i , thus the key causal links can be screened out. If $W_{:j}^t$ is zero for all t after optimization, then the information of x_j does not participate in the forecasting process of variable x_i . That means the addition of x_j cannot improve the prediction accuracy of x_i , i.e., $A_{(i,j)} = 0$. According to the GC definition, such a phenomenon illustrates that variable x_j is not the Granger cause of variable x_i and vice versa.

IV. THEORETICAL VERIFICATION AND DISCUSSIONS

The key idea of MiTCD is to adaptively recover the LCVs from DVs with the guidance of CVs, allowing causal discovery in a unified continuous-valued space. In this section, we first elaborate on the application scenarios of MiTCD and the properties of real-world mixed time series that we are concerned about (Section IV-A) and then provide theoretical explanations to answer the questions: *How does MiTCD recover the latent continuity for DVs?* (Section IV-B); *What factors does latent continuity recovery depend on and how does latent continuity recovery impact causal discovery?* (Section IV-C).

A. Application Scenarios of MiTCD

To illustrate the application scenarios of our approach and how our approach realizes latent continuity recovery and facilitates causal discovery accordingly, we first delve into the properties

Algorithm 2: Sparsity-Induced Deep Causal Relationship Learning.

Input: $\mathbf{x}^C = \{x_1, x_2, \dots, x_{p-n}\} \in \mathbb{R}^{(p-n) \times T}$,
 $\mathbf{x}^D = \{x_1, x_2, \dots, x_n\} \in \mathbb{R}^{n \times T}$, τ , λ_1 , λ_2 and λ_3 .
Output: Model parameters of Predictor $_i^C$ for each CV
($i = 1, \dots, p - n$) and Predictor $_i^D$ for each DV
($i = 1, \dots, n$).
1: Load the pre-trained parameters of $\{\text{CAGKE}_i\}_{i=1}^n$.
2: Load the pre-trained parameters of $\{\text{Decoder}_i\}_{i=1}^n$ and
froze the parameters;
3: Initialize the parameters of other models;
4: **while** not converge **do**
5: // Model Feedforward
6: **for** i in range (n) **do**
7: Recovering the LCVs for DVs
 $\tilde{x}_i^{LC} \leftarrow \text{CAGKE}_i(x_i^D)$;
8: Reconstruction of DVs $\hat{x}_i^D \leftarrow \text{Decoder}_i(\tilde{x}_i^{LC})$
9: Predict the future value of LCVs $\hat{x}_i^{LC} \leftarrow$
Predictor $_i(\{x_{j,t-\tau:t-1}^C\}_{j=1}^{p-n}; \{\tilde{x}_{j,t-\tau:t-1}^{LC}\}_{j=1}^n)$
10: **end for**
11: **for** i in range ($p - n$) **do**
12: Predict the future value of CVs $\hat{x}_i^C \leftarrow$
Predictor $_i(\{x_{j,t-\tau:t-1}^C\}_{j=1}^{p-n}; \{\tilde{x}_{j,t-\tau:t-1}^{LC}\}_{j=1}^n)$
13: **end for**
14: // Loss Calculation
15: Forecasting error loss $\mathcal{L}_{\text{fore}} \leftarrow$
 $\sum_{j=1}^{p-n} \text{MSE}(\hat{x}_{j,t}^C, x_{j,t}^C) + \sum_{j=1}^n \text{MSE}(\hat{x}_{j,t}^{LC}, \tilde{x}_{j,t}^{LC})$.
16: Reconstruction loss $\mathcal{L}_{\text{recons}} \leftarrow \sum_{j=1}^n \text{CE}(\hat{x}_{j,t}^D, x_{j,t}^D)$.
17: Sparsity loss $\mathcal{L}_{\text{spar}} \leftarrow \sum_{j=1}^p \sum_{t=1}^k \|W_{:,j}^t, \dots, W_{:,j}^k\|_2$.
18: Overall loss $\mathcal{L}_{\text{all}} \leftarrow \lambda_1 \mathcal{L}_{\text{fore}} + \lambda_2 \mathcal{L}_{\text{recons}} + \lambda_3 \mathcal{L}_{\text{spar}}$
19: // Parameter Update
20: Update the parameters of models using PGD optimizer
to minimize $\mathcal{L}_{\text{all}}$;
21: **end while**
22: **return** Optimized model parameters.

and origins of the widespread real-world mixed time series that we are concerned about.

In various application scenarios, due to measurement limitations, storage requirements, or transmission interference, some intrinsically continuous-valued signals are often recorded with discrete-valued forms and observed as discrete variables, e.g., meteorological data, product sales volume, stock returns, and health states [21], [50]. Accordingly, the observed discrete variables $\mathbf{x}^D \in \mathbb{R}^{n \times T}$ are derived from corresponding unobservable latent continuous variables $\mathbf{x}^{LC} \in \mathbb{R}^{n \times T}$ through a discretization function $h(\cdot)$ as

$$\mathbf{x}^D = h(\mathbf{x}^{LC}) \quad (16)$$

Remark 1: The discretization function $h(\cdot)$ can often be referred to as threshold-based binning, a standard signal processing technique where continuous values are quantized into distinct bins based on certain thresholds [51]. Such a function is often not injective, as $h(\cdot)$ could map distinct continuous

values to the same discrete value, causing information loss [24]. In addition, previous studies that focus on mixed data causal discovery [16], [23], [41] have also adopted the discrete variables that are quantized from latent continuous variables.

Meanwhile, some other related intrinsically continuous-valued signals can be recorded and observed as continuous variables. Under common Granger causality framework, the continuous variables $\mathbf{x}^C \in \mathbb{R}^{(p-n) \times T}$ are often causally dependent on both continuous variables $\mathbf{x}^C$ and unobservable latent continuous variables $\mathbf{x}^{LC}$, i.e., the temporal evolution of $\mathbf{x}^C$ can be formed as:

$$\mathbf{x}_t^C = g(\mathbf{x}_{t-\tau:t-1}^C, \mathbf{x}_{t-\tau:t-1}^{LC}) + \epsilon_t \quad (17)$$

where $g(\cdot)$ is the generative function of $\mathbf{x}^C$, $t-\tau:t-1$ denotes the time steps with lag τ , and ϵ_t denotes independent noises. In addition, we assume that each recording of both $\mathbf{x}^{LC}$ and $\mathbf{x}^C$ obeys Gaussian distributions.

Remark 2: The temporal evolution of continuous variables $\mathbf{x}^C$ relies on both $\mathbf{x}^C$ and $\mathbf{x}^{LC}$. On the one hand, the current values of continuous variables $\mathbf{x}_t^C$ depend on their past values due to their autoregressive nature. On the other hand, Granger causality often exists within $\mathbf{x}^C$ and between $\mathbf{x}^C$ and $\mathbf{x}^{LC}$, making the current values of continuous variables $\mathbf{x}_t^C$ depend on the past values of $\mathbf{x}^C$ and $\mathbf{x}^{LC}$. The causal relations and correlations between $\mathbf{x}^{LC}$ and $\mathbf{x}^C$ serve as the basis for achieving latent continuity recovery.

Intuitive Example: Take a typical example in the industrial system, variables such as temperature, flow rate, and pressure are often causally associated and correlated (e.g., rising temperature leads to increasing pressure and thus accelerating flow rate velocity). The temporal variations of these variables are recorded by various sensors, in which variables such as flow rate and temperature can be recorded with fine-grained continuous values, while some pressure variables may be difficult to measure directly and are usually recorded as course-grained discrete variables.

B. Recovering the LCVs With the Guidance of CVs

Essentially, recording discrete $\mathbf{x}^D$ from the underlying continuous $\mathbf{x}^{LC}$ causes information loss, thus we can not recover $\mathbf{x}^{LC}$ solely from $\mathbf{x}^D$. Studies on non-linear independent component analysis [52], [53] demonstrate that the identification of latent variables can be achieved by utilizing auxiliary variables that are statistically correlated with them. Thereby, $\mathbf{x}^C$ can provide additional information to compensate for the lost information of $\mathbf{x}^D$ during discretization, enabling the recovery of $\mathbf{x}^{LC}$. This is the key insight of MiTCD, i.e., adaptively recovering $\mathbf{x}^{LC}$ with the guidance of $\mathbf{x}^C$.

We denote the recovered LCVs as $\tilde{\mathbf{x}}^{LC}$, which are obtained by CAGKE modules that receive $\mathbf{x}^D$ as input and are learned under the guidance of $\mathbf{x}^C$. As elaborated in Section III-D, the learning objectives for latent continuity recovery involve two parts: 1) Learning to forecast future CVs $\mathbf{x}_t^C$ from the time-lagged $\mathbf{x}_{t-\tau:t-1}^C$ and $\tilde{\mathbf{x}}_{t-\tau:t-1}^{LC}$ ((8)); 2) Enforcing the reconstruction of $\mathbf{x}^D$ from $\tilde{\mathbf{x}}^{LC}$ ((9)). First, we prove that learning to forecast $\mathbf{x}_t^C$

facilitates the recovery of LCVs, as given by Theorem 1 from the perspective of information theory.

Theorem 1: Minimizing the mean square error-based forecasting loss of $\mathbf{x}_t^C$ by feeding the time-lagged $\mathbf{x}_{t-\tau:t-1}^C$ and $\tilde{\mathbf{x}}_{t-\tau:t-1}^{LC}$ (where $\tilde{\mathbf{x}}^{LC}$ is recovered from $\mathbf{x}^D$) equals to maximizing the mutual information $I(\mathbf{x}_t^C; \mathbf{x}_{t-\tau:t-1}^C, \tilde{\mathbf{x}}_{t-\tau:t-1}^{LC})$.

Proof: Please refer to Appendix A, available online.

Remark 3: The maximization of $I(\mathbf{x}_t^C; \mathbf{x}_{t-\tau:t-1}^C, \tilde{\mathbf{x}}_{t-\tau:t-1}^{LC})$ encourages the recovered $\tilde{\mathbf{x}}_{t-\tau:t-1}^{LC}$ to contain as much information about $\mathbf{x}_t^C$ as possible. Consequently, latent continuity recovery can restore the information of the ground-truth $\mathbf{x}^{LC}$ that are correlated with $\mathbf{x}^C$.

On top of this, the second objective guarantees that the recovered $\tilde{\mathbf{x}}^{LC}$ can preserve the information of discrete variables $\mathbf{x}^D$. This serves as a regularization for latent continuity recovery, enforcing the recovered $\tilde{\mathbf{x}}^{LC}$ to include relevant information about actual $\mathbf{x}^{LC}$ and exclude unnecessary noises. Consequently, these two objectives work collaboratively to achieve accurate latent continuity recovery. Our visualization plots in Fig. 10 further verify that our approach can accurately recover the latent continuous variables on various datasets with diverse temporal dynamics.

C. Feasibility and Impact of Latent Continuity Recovery

Feasibility of Latent Continuity Recovery: As discussed in Remark 3, latent continuity recovery can restore the correlated information of $\mathbf{x}^{LC}$ with $\mathbf{x}^C$. Thereby, the feasibility of latent continuity depends on the amount of information $\mathbf{x}^C$ can provide, which may be compromised in some cases where CVs are not associated with LCVs or cannot provide sufficient information for accurate recovery of LCVs. Take a typical example where most of the observed variables are DVs and only a few are CVs, it may be difficult for some DVs to obtain sufficient information for recovering their LCVs.

Impacts on Causal Discovery: Through reliable latent continuity recovery, we can bridge the information granularity gap between DVs and CVs and unify the distribution types of mixed variables, allowing more flexible and reliable causal discovery in a unified continuous-valued space. Our experiments, particularly the portability ability analysis in Section V-C2 further demonstrate that our recovered LCVs can also benefit existing causal discovery methods, e.g., LGC [9] and cMLP [12], to promote their performance in mixed time series causal discovery tasks.

In addition, we study the impact on causal discovery when the feasibility of latent continuity recovery may be compromised. Specifically, our empirical studies in Section V-B5 further investigate the effect of the proportions of DVs on causal discovery. The results demonstrate that *with the increase in the proportion of DVs to the whole variables, the latent continuity recovery becomes difficult, and thus inevitably reduces the causal discovery accuracy*. In the real world, however, mixed time series primarily consists of CVs, particularly in domains such as finance, healthcare, and industrial systems [27]. The prevalence of CVs supports the latent continuity recovery to be feasible and reliable, enabling accurate causal discovery for mixed time series.

TABLE II
SUMMARY OF DATASETS

Dataset	Setting	Variable Dim.			Causal Sparsity	Series Length	Semantic Information Description
		Sum	CVs	DVs			
VAR	I	10	7	3	20.00%	200	Numerical Linear Lagged Regression $lag = 3$
	II	10	7	3	20.00%	500	
	III	10	7	3	20.00%	1000	
	IV	10	7	3	20.00%	2000	
Lorenz-96	I	10	6	4	40.00%	1000	Climate Science Interactions $F = \{5, 10, 15\}$
	II	15	9	6	26.67%	1000	
	III	20	12	8	20.00%	1000	
fMRI	I	10	6	4	21.00%	1200	Brain Connections of different Spatial Regions
	II	10	5	5	21.00%	1200	
MS-LotVol.	I	12	7	5	25.00%	1000	Biological Population Dynamic Systems
	II	12	6	6	25.00%	1000	
Dream3-challenge	E.C.1	100	80	20	1.25 %	950	Gene Expression Network of Silico Signals
	E.C.2	100	80	20	1.19 %	950	
	Y.S.1	100	80	20	1.66 %	950	
	Y.S.2	100	80	20	3.89 %	950	
	Y.S.3	100	80	20	5.51 %	950	

Kindly Note: "Causal sparsity" represents the proportion of causal connections in the summary causal graph. Min-max normalization is adopted to preprocess the data into the range between 0 and 1. Besides, DVs are randomly selected and are obtained with the discretization of the threshold value 0.5.

V. EXPERIMENTS

In this section, we conduct extensive experiments to evaluate the performance of MiTCD with 18 well-acknowledged and advanced temporal causal discovery methods. The experiments are carried out on five widely adopted benchmark datasets with different settings. The datasets emerge with different temporal dynamics coming from various applications and cover varying 1) *variable dimensions*; 2) *discrete variable proportions*; 3) *causal sparsity*; 4) *amount of training samples* and 5) *chaotic dynamic degrees* to comprehensively evaluate the MiTS causal discovery performance. Furthermore, we conduct abundant dataset sensitivity experiments, ablation studies, model parameter sensitivity analysis, visualization examples, and portability experiments with in-depth analysis for fine-scale evaluation and investigation of our proposed MiTCD from diverse perspectives.

A. Experimental Configurations

In this subsection, the datasets, experimental settings, and baseline methods are presented. Besides, more detailed information can be found in Appendix B, C, and E, available online.

1) *Dataset and Experimental Setting:* We summarize the properties and statistics of benchmark datasets in Table II. In the main experiments, the discrete variables in mixed time series are set as *binary* discrete variables, as they are most commonly encountered in the real world and are the most challenging ones for latent continuity and causal discovery compared with those with more discrete states. The binary discrete variable setting is consistent with previous work [16], [23]. In addition, we further analyze the discrete variables with different numbers of discrete states in Section V-C5. Here are the descriptions and settings of the datasets:

Vector Autoregression (VAR): To analyze and compare the performance under linear dynamic causally coupled processes, VAR datasets are simulated and adopted. Specifically, we set $p = 10$, $n = 3$, $lag = 3$, and different lengths of $T = \{200, 500, 1000, 2000\}$ to analyze the performance under different sample amounts.

Lorenz-96 System: To verify the nonlinear temporal causal discovery accuracy, the Lorenz-96 system is adopted. The Lorenz-96 system is a typical chaotic dynamic system, which has been extensively used for climate science with modeling and prediction tasks [54]. It is also a standard GC discovery benchmark [12], [13]. In this experiment, we set $T = 1000$, and we consider different settings of varying dimension (p and n); and varying chaotic dynamic degrees (F) as presented in Table II for performance evaluation.

Functional Magnetic Resonance Imaging (fMRI): Functional magnetic resonance imaging (fMRI) forward model consists of realistic simulation of blood-oxygen-level-dependent (BOLD) time series [55] with nonlinearly coupled causal structures. In this experiment, we choose the sixth dataset with $p = 10$ and $T = 1200$. We set different values of DVs $n = \{5, 6\}$ to consider varying discrete proportions (n/p) and evaluate the corresponding experimental results.

Multiple Species LotKa-Volterra (MS-LotVol.): Multiple Species LotKa-Volterra is a typical population dynamic system, which consists of multiple original two-species systems [56]. The original two-species systems include the population of one predator and the population of one prey species. We adopted the official implementation to obtain time series with $p = 12$ and $T = 1000$. We set different values of DVs $n = \{4, 5\}$ to consider varying discrete proportions (n/p) and evaluate the corresponding experimental results.

Dream3-challenge: Dream3-challenge [57] consists of five realistic time course gene expression sub-datasets, each with different ground truth Granger causality graphs and different causal sparsity. Compared with the above datasets, Dream3 datasets are more challenging due to the high dimensionality of variables ($p = 100$), limited data scenario relative to the dimensionality, and very sparse causal structure (1.19%~5.51%).

2) Baseline Models and Evaluation Metrics: Baselines and Implementations: For a comprehensive evaluation, we compare MiTCD with well-acknowledged and advanced temporal causality discovery methods, including Lasso Granger Causality (LGC)¹ [9] (2007), VARLiNGAM² [31] (2010), PCMCi³ [6] (2019), TCDF⁴, eSRU⁵ [13] (2020), GVAR⁶ [14] (2021), cMLP and cLSTM⁷ [12] (2022), Hetcor-PC⁸ [40] (2019), Dynotears⁹ [33] (2020), NTSnotears¹⁰ [34] (2023), SS-CASTLE¹¹ [32]

(2023), MS-CASTLE¹² [32] (2023), PCTMI¹³ [36] (2022), NTiCD¹⁴ [29] (2023), CR-VAE¹⁵ [15] (2023), and RegCI¹⁶ [41] (2023).

All methods are implemented according to their official codebase and settings. We implemented general PCMCi algorithm [6] with two typical conditional independent (CI) test approaches, including *nearest-neighbor conditional mutual information test* [58] (denoted by PCMCi^C), and *partial correlation independent test* (denoted by PCMCi^P). Besides, RegCI [41] and Hetcor-PC [40] are designed as CI methods for mixed data, we incorporated them with the PCMCi framework for implementation.

The experimental platform is a server with Intel(R) Xeon(R) Gold 6246 CPUs and NVIDIA 2080Ti GPUs. The versions of Python, Scikit-learn, and Pytorch are 3.9.7, 1.0.1, and 1.10.0 respectively.

Evaluation Metrics and Hyper-parameter Settings: To measure the causal discovery performance, areas under receiver operating characteristic (AUROC) and precision-recall (AUPRC) curves as evaluation metrics in accordance with previous works [12], [13], [14]. To ensure fair and comprehensive evaluations, we extensively tune the hyper-parameters of each baseline method to obtain the best AUPRC score.¹⁷ The detailed introductions of baseline methods and the configurations of hyper-parameters can be found in Appendix B and D, available online, respectively.

B. Experimental Results and Discussions

In this subsection, the overview of experimental results, performance comparison of baseline methods, and dataset sensitivity analysis are presented with quantitative tables and corresponding visualization results and discussions.

1) Main Results Overview and Comparison: Fig. 4 concludes the main results of baseline methods and our MiTCD on five causal discovery benchmark datasets. The quantitative results of VAR dataset, Lorenz-96 system, fMRI and MS-Lotvol. datasets, and Dream-3 challenge are summarized in Tables IV, III, V, and VI, respectively. In general, as presented in Fig. 4, MiTCD achieves consistent state-of-the-art performance on five main datasets with MiTS. Here are the specific discussions and analyses for experiments on each dataset.

i) **Results on VAR datasets:** Since VAR datasets are generated from linear dynamic processes, the causal relation among MiTS is relatively easy to identify compared with other datasets. From Table IV, we can observe that most baseline methods achieve feasible results. Among them, MiTCD presents the best performance, which can obtain

¹https://github.com/cloud36/graphical_granger_methods

²https://github.com/ckassaad/causal_discovery_for_time_series

³<https://github.com/jakobrunge/tigramite>

⁴<https://github.com/M-Nauta/TCDF> [28] (2019)

⁵https://github.com/sakhanna/SRU_for_GCI

⁶<https://github.com/i6092467/GVAR>

⁷<https://github.com/iancovert/Neural-GC>

⁸<https://github.com/cuiruifei/HetcorPC>

⁹https://github.com/ckassaad/causal_discovery_for_time_series

¹⁰<https://github.com/xiangyu-sun-789/NTS-NOTEARS>

¹¹<https://github.com/OfficiallyDAC/mcsl>

¹²<https://github.com/OfficiallyDAC/mcsl>

¹³<https://github.com/ckassaad/PCTMI>

¹⁴<https://github.com/SaimaAbsar/NTiCD>

¹⁵<https://github.com/hongmingli1995/CR-VAE>

¹⁶<https://github.com/jakobrunge/tigramite>

¹⁷Since some baseline methods can not directly handle MiTS with DVs, we make modifications on the decoders and loss functions. Concretely, mean square error (MSE) loss is used for forecasting CVs; {ReLU activation + MSE loss} or {logistic activation + cross-entropy (CE) loss} are used for forecasting of DVs, depending on the grid search results of the final performance.

TABLE III
PERFORMANCE COMPARISON ON LORENZ-96 SYSTEMS

Method	$p = 10, n = 4$						$p = 15, n = 6$						$p = 20, n = 8$					
	$F = 5$		$F = 10$		$F = 15$		$F = 5$		$F = 10$		$F = 15$		$F = 5$		$F = 10$		$F = 15$	
	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC
LGC	59.58	46.07	62.50	48.57	60.83	47.02	61.52	33.72	64.47	36.76	65.98	38.81	67.66	33.44	68.13	33.81	66.56	31.99
VARLiNGAM	77.50	63.83	87.50	77.33	92.08	84.48	81.52	57.60	90.38	77.54	90.76	76.34	87.66	61.25	91.19	70.22	91.72	72.26
PCMCI ^C	59.63	45.55	76.13	60.25	80.33	63.86	74.74	46.68	84.09	57.45	81.32	55.44	76.72	41.60	89.16	61.85	87.23	57.38
PCMCI ^P	51.60	41.21	55.71	43.76	63.38	49.61	62.32	32.88	73.09	41.10	77.76	45.74	80.52	41.23	82.94	43.27	82.64	42.63
TCDF	53.67	49.46	52.21	42.44	50.56	42.03	50.27	30.69	49.38	29.16	49.68	28.91	50.34	22.51	50.06	20.73	50.18	20.34
eSRU	87.08	85.70	88.92	88.70	93.38	92.00	86.35	77.32	95.12	92.02	93.90	90.63	95.19	88.27	92.95	87.44	93.21	<u>88.68</u>
GVAR	73.08	68.02	71.33	68.52	72.38	66.70	73.02	57.08	75.54	59.96	75.19	60.09	72.54	56.71	69.46	54.29	67.41	57.05
cMLP	91.08	89.31	<u>95.69</u>	<u>94.88</u>	<u>96.85</u>	<u>95.75</u>	89.56	84.60	92.93	89.39	<u>95.70</u>	<u>93.23</u>	<u>98.82</u>	91.66	93.30	88.93	87.10	79.35
cLSTM	90.50	<u>91.51</u>	90.79	90.02	94.92	94.11	<u>91.27</u>	<u>87.98</u>	<u>97.40</u>	<u>94.93</u>	93.75	90.20	97.93	<u>94.53</u>	95.21	88.05	93.01	83.66
Hetcor-PC	65.42	55.89	74.31	57.92	79.52	64.66	66.31	49.75	72.56	47.92	86.56	62.31	84.23	51.23	87.93	54.22	89.12	55.23
Dynotears	69.17	57.04	72.50	59.64	69.92	54.56	69.32	44.93	93.79	50.88	72.05	44.04	78.44	47.16	73.79	50.88	75.78	42.75
NTSnotears	82.04	77.19	84.04	75.45	87.85	81.82	87.92	79.48	86.80	73.55	88.85	74.43	88.73	74.86	89.45	74.18	93.38	76.72
SS-CASTLE	61.60	56.84	66.21	60.11	71.25	65.50	71.27	58.14	68.14	53.56	67.29	52.26	74.38	59.00	64.89	44.00	67.91	48.87
MS-CASTLE	63.33	62.69	60.67	62.09	60.29	61.06	70.16	57.27	67.12	54.24	65.92	52.33	76.28	60.12	67.18	47.88	67.29	47.96
PCTMI	74.21	58.27	74.37	57.57	77.17	60.10	80.18	55.41	88.71	62.30	84.64	57.65	86.31	65.43	82.88	66.09	84.20	59.41
NTiCD	76.08	77.43	76.63	78.49	69.67	59.94	69.67	61.40	69.39	62.17	70.90	62.21	69.34	55.34	70.86	56.75	69.56	54.50
CR-VAE	89.56	88.30	91.08	90.89	89.75	88.76	88.41	81.03	96.02	92.32	92.47	86.24	96.96	92.28	<u>96.89</u>	<u>91.47</u>	<u>94.82</u>	87.13
RegCI	45.98	48.12	64.88	49.15	74.33	57.19	65.17	35.22	76.17	44.79	82.72	53.14	77.20	38.19	83.10	44.29	88.29	53.15
MiTCD	96.88	96.02	97.50	97.00	98.46	98.00	95.86	92.07	99.64	99.16	96.59	93.78	99.98	99.91	99.69	98.84	96.14	91.74
$\Delta \text{ imp.}$	$\uparrow 5.80$	$\uparrow 4.51$	$\uparrow 1.81$	$\uparrow 2.12$	$\uparrow 1.61$	$\uparrow 2.25$	$\uparrow 4.59$	$\uparrow 4.09$	$\uparrow 2.24$	$\uparrow 4.23$	$\uparrow 0.89$	$\uparrow 0.55$	$\uparrow 1.16$	$\uparrow 5.38$	$\uparrow 2.80$	$\uparrow 7.37$	$\uparrow 1.32$	$\uparrow 3.06$

Kindly Note: PCMCI^C and PCMCI^P denote PCMCI with nearest-neighbor conditional mutual information test and partial correlation independent test, respectively. The best results are highlighted in **bold** and the second best results are highlighted with underline. “ $\Delta \text{ imp.}$ ” denotes the performance improvement compared to the best baseline method. The same for Table 4, 5, and 6.

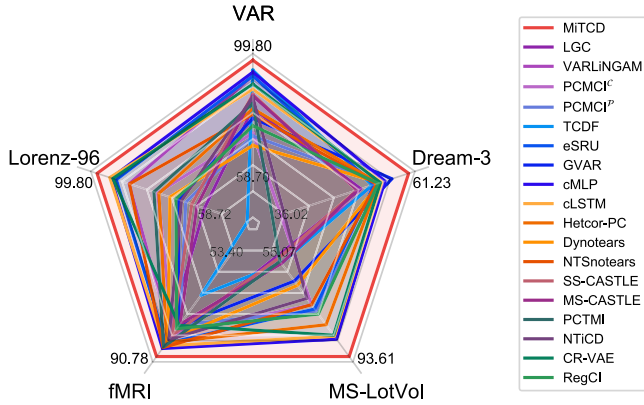


Fig. 4. Overall MiTS causal discovery performance comparison of baseline methods and our MiTCD on five main datasets. Kindly note: the results are evaluated by the AUROC scores (%) that are averaged from the sub-datasets.

performance improvements of up to 5.87% of AUROC and 8.38% of AUPRC scores compared with the second best methods TCDF and NTiCD, respectively.

- ii) *Results on Lorenz-96 systems:* Lorenz-96 systems are typical nonlinear systems with complex chaotic dynamics. As listed in Table III, some linear methods, e.g., LGC and RegCI report poor results, while other deep learning-based causal discovery methods, e.g., cMLP, cLSTM and CR-VAE create satisfactory performance. Besides, MiTCD further presents more superior performance, which impressively obtaining 5.80% in AUROC improvement and 4.51% in AUPRC improvement when

TABLE IV
PERFORMANCE COMPARISON ON VAR SIMULATION

Method	$T = 200$		$T = 500$		$T = 1000$		$T = 2000$	
	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC
LGC	78.75	37.04	81.88	42.02	86.88	48.78	86.25	49.78
VARLiNGAM	87.50	68.00	93.12	71.33	94.38	87.27	97.50	83.33
PCMCI ^C	70.09	37.20	75.25	41.91	80.56	50.33	78.47	45.83
PCMCI ^P	72.06	39.10	71.50	39.83	74.50	43.44	82.88	51.33
TCDF	<u>88.25</u>	56.95	93.25	<u>85.58</u>	<u>98.62</u>	94.42	<u>99.18</u>	<u>96.98</u>
eSRU	87.06	68.99	92.93	85.30	97.44	92.59	95.19	88.57
GVAR	67.88	41.58	74.62	46.64	90.38	73.05	88.94	73.72
cMLP	87.03	67.45	<u>94.63</u>	84.71	97.37	<u>96.00</u>	98.06	95.81
cLSTM	72.12	54.77	89.91	67.31	97.31	95.32	97.50	93.22
Hetcor-PC	81.23	53.12	83.22	56.33	83.49	57.12	85.56	59.13
Dynotears	67.50	29.84	72.50	34.82	71.25	34.26	78.32	44.13
NTSnotears	77.04	58.09	83.25	71.00	83.81	60.09	82.75	59.83
SS-CASTLE	86.91	77.24	87.78	80.00	87.50	80.00	88.31	83.42
MS-CASTLE	86.19	76.11	86.93	77.33	87.97	78.79	87.72	77.22
PCTMI	79.94	55.00	80.59	45.49	90.63	62.50	88.50	60.00
NTiCD	86.88	<u>76.01</u>	85.00	76.43	87.10	78.12	88.21	78.97
CR-VAE	82.56	64.20	90.50	67.72	93.94	83.64	95.81	86.86
RegCI	76.63	44.60	78.40	48.21	78.12	47.49	79.94	50.31
MiTCD	94.12	84.39	97.38	93.56	99.87	99.55	100.00	100.00
$\Delta \text{ imp.}$	$\uparrow 5.87$	$\uparrow 8.38$	$\uparrow 2.75$	$\uparrow 7.98$	$\uparrow 1.25$	$\uparrow 3.55$	$\uparrow 0.82$	$\uparrow 3.02$

$F = 5$ and $p = 10$ compared to the second-best methods cMLP and cLSTM, respectively. Besides, MiTCD achieves 2.80% in AUROC and 7.37% in AUPRC improvement in high-dimensional scenarios ($F = 10$ and $p = 20$) compared to the second-best methods CR-VAE.

TABLE V
PERFORMANCE COMPARISON ON fMRI AND MS-LotVol

Method	fMRI				MS. LotKa-Volterra			
	$n = 4$		$n = 5$		$n = 5$		$n = 6$	
	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC
LGC	84.15	54.97	85.46	54.29	57.87	35.12	56.48	33.10
VARLiNGAM	83.18	62.39	81.43	61.90	59.31	34.88	56.42	33.54
PCMCIC	82.91	55.14	80.11	51.08	77.52	64.31	70.21	55.12
PCMCIP	78.09	42.45	78.57	41.28	78.13	62.75	73.97	56.57
TCDF	84.87	63.17	51.99	24.23	64.09	42.17	54.89	32.00
eSRU	86.14	74.83	83.06	74.09	77.11	60.10	73.53	58.20
GVAR	80.23	66.91	77.40	66.29	66.38	53.25	64.87	50.21
cMLP	86.98	<u>76.72</u>	<u>85.62</u>	<u>74.49</u>	<u>86.37</u>	72.93	<u>85.15</u>	<u>71.99</u>
cLSTM	85.59	70.96	85.23	73.33	85.35	<u>75.01</u>	84.16	69.93
Hetcor-PC	84.35	55.63	84.23	52.13	83.22	69.38	78.32	64.23
Dynotears	81.62	48.39	85.74	52.56	68.06	49.50	66.20	43.59
NTSnotears	<u>87.76</u>	72.98	84.18	67.65	75.62	60.65	72.02	54.85
SS-CASTLE	84.63	68.41	78.96	57.25	58.33	35.14	57.44	34.13
MS-CASTLE	76.19	68.92	80.11	71.82	58.87	38.27	60.24	39.80
PCTMI	85.90	62.10	85.62	60.05	58.68	28.90	58.92	29.43
NTiCD	83.24	71.65	83.42	72.69	72.09	64.68	71.19	60.96
CR-VAE	79.81	63.91	77.28	63.84	84.85	71.66	83.95	69.45
RegCI	79.23	43.74	80.11	45.54	79.17	68.95	75.00	62.50
MiTCD	90.36	80.88	87.64	78.19	91.29	77.69	92.26	78.28
$\Delta imp.$	$\uparrow 2.60$	$\uparrow 4.16$	$\uparrow 2.02$	$\uparrow 3.70$	$\uparrow 4.92$	$\uparrow 2.68$	$\uparrow 7.11$	$\uparrow 6.29$

TABLE VI
PERFORMANCE COMPARISON ON DREAM3-CHALLENGE

Method	Elico.1	Elico.2	Yeast.1	Yeast.2	Yeast.3	Average
	AUROC	AUROC	AUROC	AUROC	AUROC	AUROC
LGC	52.41	49.42	51.83	50.88	49.66	50.84
VARLiNGAM	51.79	57.07	53.69	51.40	52.81	54.53
PCMCIP	57.37	54.52	56.44	53.03	51.27	54.53
TCDF	49.14	58.19	52.57	52.81	52.20	52.98
eSRU	55.89	51.43	53.73	52.38	<u>53.35</u>	53.36
GVAR	<u>60.59</u>	<u>60.79</u>	<u>58.22</u>	<u>54.74</u>	49.72	<u>56.81</u>
cMLP	59.90	54.03	57.06	53.71	52.61	55.46
cLSTM	57.96	53.90	57.42	52.73	52.44	54.89
Hetcor-PC	56.38	54.23	54.12	54.57	51.93	54.25
Dynotears	56.18	57.33	56.32	51.43	51.06	54.46
NTSnotears	56.71	51.50	57.11	51.18	52.40	53.78
SS-CASTLE	50.22	50.00	50.14	50.02	50.03	50.08
MS-CASTLE	50.48	50.21	50.10	50.02	50.00	50.16
CR-VAE	55.38	54.12	57.84	54.23	51.29	54.57
RegCI	55.96	55.53	55.25	52.69	51.40	54.17
MiTCD	64.53	63.23	59.42	58.01	54.96	60.03
$\Delta imp.$	$\uparrow 3.94$	$\uparrow 2.44$	$\uparrow 1.20$	$\uparrow 3.27$	$\uparrow 1.61$	$\uparrow 3.22$

Kindly note: Due to the limitation of computational effort, the results of PCMCIC, PCTMI and NTiCD are not reported in Dream3 experiments.

The experimental results demonstrate the superior causal discovery ability of MiTCD in multivariate nonlinear chaotic scenarios.

- iii) *Results on fMRI and MS-LotVol. Datasets:* In the experiments on fMRI and MS-LotVol. datasets, we consider the settings of different DV proportions. The results are reported in Table V. In line with our expectations, it becomes more challenging when the proportion of DVs increases. Especially for TCDF on fMRI, when $n = 4$

becomes $n = 5$, the AUROC score decreases by 32.88% (84.87% $\rightarrow$ 51.99%); for TCDF on MS.LotKa-Volterra, the AUPRC score decreases by 12.17% (42.17% $\rightarrow$ 32.00%). Surprisingly, LGC presents competitive results on fMRI, i.e., obtaining an AUROC score of 84.15% when $n = 4$ and an AUROC score of 85.46% when $n = 5$. In summary, MiTCD attains the best performance across different settings and outperforms the baseline methods by a large margin, e.g., increasing the AUPRC score by 4.16% on fMRI when $n = 4$, and increasing the AUROC score on MS. LotKa-Volterra by 7.11% on fMRI when $n = 6$.

- iv) *Results on Dream3-challenge:* As aforementioned, inferring silicon network from dream-3 datasets is very challenging, especially in the context of mixed data. From Table VI, we can observe that eSRU and GVAR obtain competitive results on part of the sub-datasets. Overall, MiTCD outperforms all baselines across all of the five sub-datasets and improves the average AUROC score by 3.22%. Especially for Elico.1, MiTCD improves the AUROC score by 3.94% compared with the second-best baseline GVAR. Besides, the raw receiver operating characteristics (ROC) curves of MiTCD on five sub-datasets are displayed in Fig. 6, which represents the trade-off between the true-positive rate (TPR) and the false-positive rate (FPR) achieved by MiTCD.

In summary, the linear method LGC and VARLiNGAM obtain good performance on VAR and fMRI datasets. Besides, each of the deep causal discovery methods, such as TCDF, GVAR, cMLP, cLSTM, and NTSnotears presents competitive results on different datasets. However, as they are mostly developed for continuous time series, their performance would inevitably decay when encountering MiTS. Fortunately, empowered by the flexible CAGKE technique and corresponding training strategies, our MiTCD can address the InfoA problem and achieve superior performance for MiTS causal discovery in the unified continuous-valued space.

- 2) *Causal Discovery Results Visualization:* To provide a better understanding of causal discovery results, Fig. 5 presents the ground truth of causal summary graphs of VAR, Lorenz-96, fMRI and MS-LotVol datasets as well as the estimation results of MiTCD. From Fig. 5, we can notice that different datasets contain different causal structure patterns and causal sparsity degrees since they characterize interaction information of different semantic domains. For example, the causal relationships of VAR datasets are randomly scattered, while the causal relationships of Lorenz-96 and MS-LotVol exhibit regular patterns. The visualization results demonstrate the excellent causal discovery ability of MiTCD on different datasets, as it can accurately reveal the true causal structures in accordance with the ground truth. For datasets with linear and simple causal structures, MiTCD can accurately capture the interactions of mixed variables through sparse-induced causal structure learning. For complex nonlinear datasets, MiTCD can also reconstruct the latent continuity information of DVs and model the causal relations via self-supervised pre-trained CAGKE and deep causal learning.

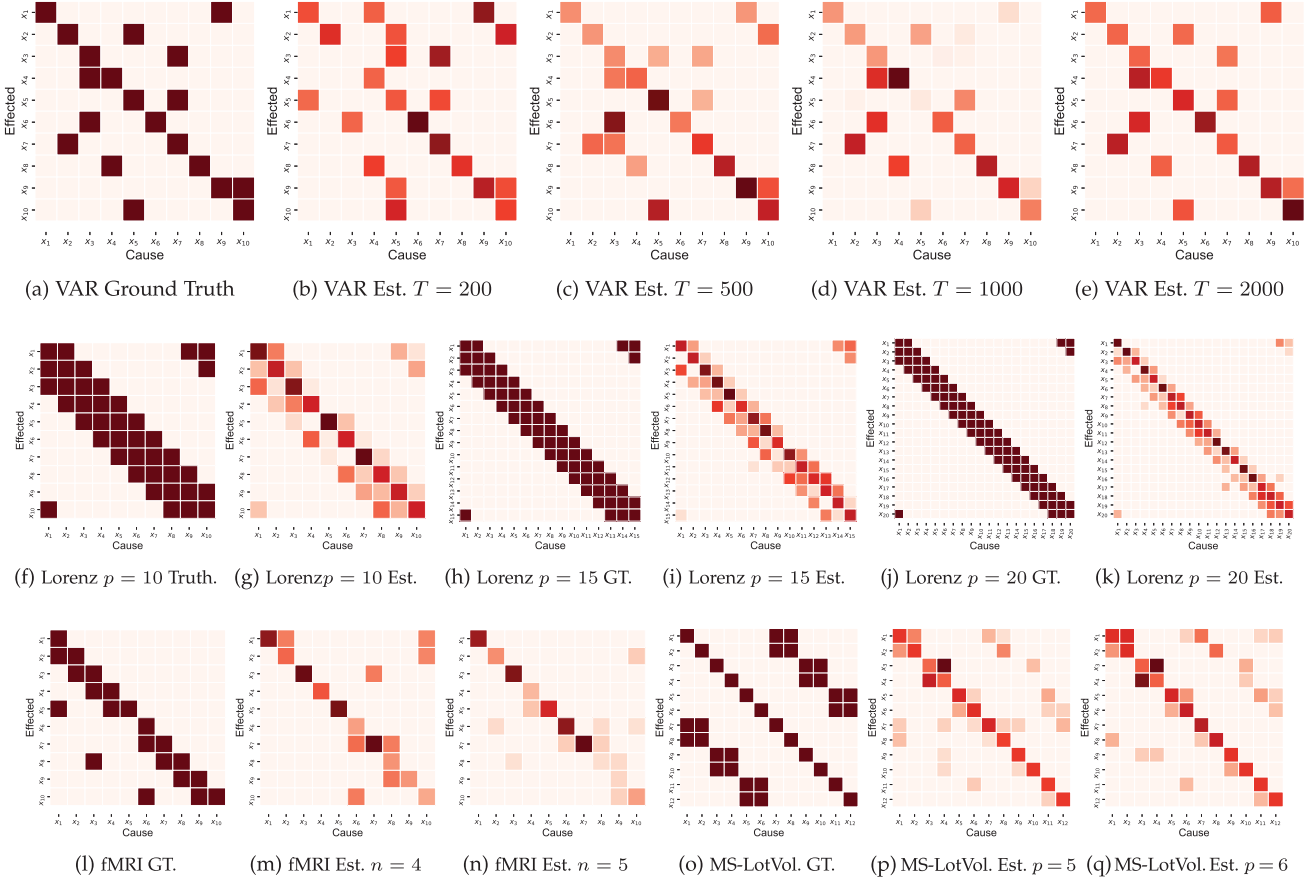


Fig. 5. Summary causal graphs visualization of the ground truths and estimation results obtained by MiTCD. Kindly note: for each ground truth causal summary graph, Dark red cells correspond to the existence of GC relationships; light red cells denote the absence of the GC relationships. The VAR datasets with varying T share the same causal structure. The fMRI and MS-LotVol. datasets with varying n also share the same causal structure.

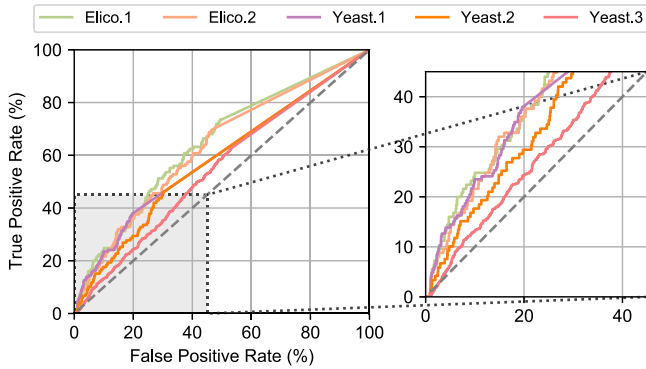


Fig. 6. ROC curves of Dream-3 challenge.

3) *Sensitivity Analysis of Training Data Amount:* The amount of training data is a key factor that impacts the accuracy of causal discovery. To further investigate its influence, we conduct experiments with different training data lengths T on the VAR dataset. Specifically, the quantitative results are summarized in Table IV. Besides, a corresponding visualization plot is shown in Fig. 7 to provide more conceptual understanding, from which we can observe that most methods achieve good

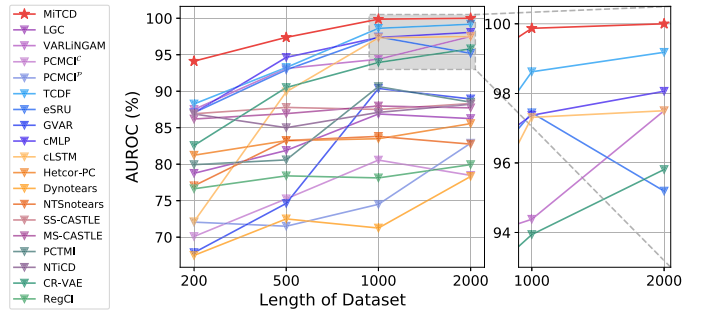


Fig. 7. Data amount sensitivity analysis (based on VAR datasets).

performance when the training data is abundant. However, the performance of baseline methods degenerates terribly with the decline of training data amount. For example, when the training data length decreases from 1000 to 200, the performance of GVAR declines 31.47% in AUPRC score (73.05% $\rightarrow$ 41.58%), the performance of cLSTM declines 25.19% in AUROC score (97.31% $\rightarrow$ 72.12%). However, benefiting from the flexible representation ability of pre-trained CAGKE and sparsity-induced optimization techniques, our MiTCD still realizes remarkable

TABLE VII
ABLATION STUDY RESULTS OF MITCD AND ITS VARIANTS

Model	$\mathcal{L}_{rec}$	$\mathcal{L}_{all}^{pre}$	CAGKE	VAR $T = 1000$				Lorenz-96 $p = 10, F = 5$				fMRI $n = 4$				MS. LotKa-Volterra $n = 5$			
				AUROC	Δ	AUPRC	Δ	AUROC	Δ	AUPRC	Δ	AUROC	Δ	AUPRC	Δ	AUROC	Δ	AUPRC	Δ
MiTCD	✓	✓	✓	99.87	-	99.55	-	96.88	-	96.02	-	90.36	-	80.88	-	91.29	-	77.69	-
MiTCD [†]	✗	✓	✓	99.50	↓0.37	98.26	↓1.29	95.38	↓1.50	93.84	↓2.18	90.10	↓0.26	80.72	↓0.16	91.49	↑0.20	76.04	↓1.65
MiTCD [‡]	✗	✗	✓	99.31	↓0.56	97.46	↓2.09	94.12	↓2.76	91.75	↓4.27	87.58	↓2.78	79.63	↓1.25	86.90	↓4.39	72.13	↓5.56
MiTCD [§]	✗	✗	✗	97.03	↓2.84	94.66	↓4.89	92.11	↓4.77	88.12	↓7.90	86.44	↓3.92	76.01	↓2.87	86.84	↓4.45	73.12	↓4.57

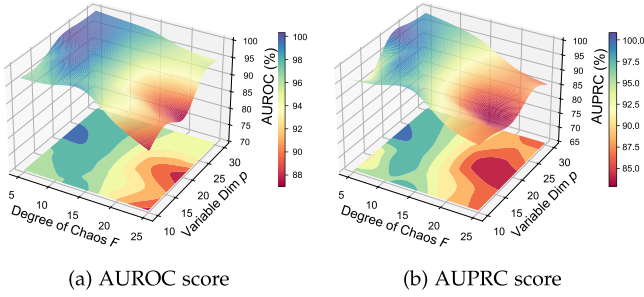


Fig. 8. Causal discovery sensitivity analysis of variable dimension (p) and chaos degree (F) of Lorenz-96 systems.

causal discovery performance even under small T . Besides, MiTCD outperforms baseline methods under all settings with varying training data lengths.

4) *Sensitivity Analysis of Variable Dimension and Chaotic Dynamic Degrees*: In Lorenz-96 systems, the variable dimension p and the chaotic dynamic degree F are key setting parameters that affect the interactions among variables. In this part, we conduct sensitivity experiments under different $p = \{10, 15, 20, 25, 30\}$ and different $F = \{5, 10, 15, 20, 25\}$ and investigate the concurrent impacts to causal discovery performance. For fair evaluation, we keep the proportion of DVs ($n/p = 40\%$) equivalent for all settings. The experimental results are shown in Fig. 8. Accordingly, we have observations as follows: 1) Best performance is obtained when $p = 20$ and $F = 5$, which may be attributed to the causal sparsity degree. As shown in Fig. 5, when p is small, the causal structure is relatively dense. When p is large, the causal structure becomes relatively sparse. Technically, both too dense and too sparse causal structures are hard to identify. Thereby, the best performance would be potentially obtained in a proper sparsity degree; 2). Generally, when the chaotic dynamic degree F increases, the causal discovery performance decreases. It is because with the increase of F , the nonlinearity of the interactions between variables becomes larger, as well as more noise is introduced, making the time series more unpredictable and making the identification of causal structures more difficult.

5) *Varying Discrete Variable Proportion Analysis*: In this part, we investigate the impact of proportions of DVs based on Lorenz-96 systems across a relatively large range ($n/p = 10\% \sim 80\%$) and summarize the results in Table VIII and visualize the results in Fig. 9. It can be noticed that as the proportions of DVs increase, the accuracy of causal discovery decreases. In the range of (10% ~ 40%), the performance remains at a high level (i.e.,

TABLE VIII
PERFORMANCE ANALYSIS ON VARYING DISCRETE VARIABLE PROPORTIONS

DVs	$n = 1$	$n = 2$	$n = 3$	$n = 4$	$n = 5$	$n = 6$	$n = 7$	$n = 8$
AUROC	99.29	98.75	97.17	97.50	94.56	94.13	90.08	86.46
AUPRC	99.00	98.17	96.50	97.00	93.15	92.30	88.11	84.60

Kindly Note: The experiment is based on Lorenz=96 system ($p = 10, F = 10$).

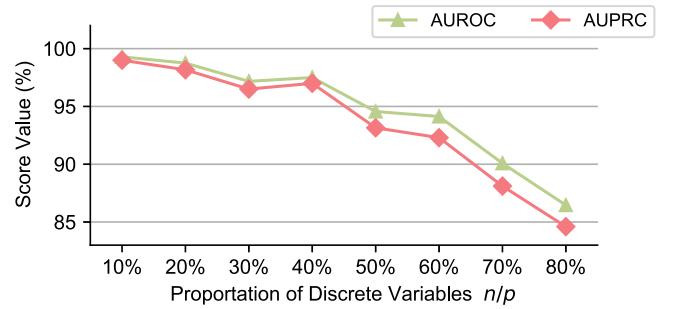


Fig. 9. Discrete Variable Proportion sensitivity analysis.

$\text{AUROC} \geq 95\%$), but it drops significantly when n/p above 50%. These results are in line with our expectations, as the introduction of more DVs decreases the overall information density. Besides, the reduction of CVs also reduces the supervisory information for optimal parameter learning of CAGKE, making it a struggle to recover the latent continuity of DVS. Consequently, the causal discovery performance decays.

C. Model Investigation and In-Depth Analysis

1) *Model Ablation Analysis*: To further verify the effectiveness of each specific design of MiTCD, we conduct ablation experiments to compare MiTCD with its several model variants: 1). ablating the discrete variable reconstruction loss $\mathcal{L}_{rec}$ (denoted by MiTCD[†]); 2). ablating the pre-training stage (denoted by MiTCD[‡]); 3). ablating the CAGKE and adopts a single and unlearnable Gaussian kernel embedding ($\sigma = 1$) to obtain LCVs (denoted by MiTCD[§]).

We conduct ablation experiments on both VAR, Lorenz-96, fMRI and MS. LotKa-Volterra datasets. Accordingly, the results are summarized in Table VII, from which we can draw the following observations and results: 1) Removing the $\mathcal{L}_{rec}$ slightly decreases the overall performance, which means that the discrete variable reconstruction loss can help to constraint the LCV recovering process and is beneficial for causal structure learning; 2) pre-training stage of CAGKE could enhance the

TABLE IX
PORTABILITY EXPERIMENTS OF PRE-TRAINED CAGKE

Method	AUROC	Δ AUROC	AUPRC	Δ AUPRC	Time (s)	Δ Time(s)
LGC	59.58	-	47.06	-	1.2	-
+CAGKE	63.44	$\uparrow 3.86$	50.96	$\uparrow 3.90$	6.8	+5.6
GVAR	73.08	-	68.02	-	13	-
+CAGKE	78.85	$\uparrow 5.77$	75.21	$\uparrow 7.19$	19	+5.6
eSRU	87.08	-	87.50	-	117	-
+CAGKE	88.17	$\uparrow 1.09$	88.33	$\uparrow 0.83$	123	+5.6
cMLP	91.08	-	89.31	-	1719	-
+CAGKE	93.96	$\uparrow 2.88$	92.80	$\uparrow 3.49$	1724	+5.6
cLSTM	90.50	-	91.51	-	1839	-
+CAGKE	95.53	$\uparrow 5.03$	94.62	$\uparrow 3.11$	1845	+5.6

Kindly Note: The experiment is based on Lorenz-96 system ($p = 10, F = 5$).

model expressive capacity thus facilitating the causal discovery. Empowered by pre-training, the AUROC score can be improved by 2.78% on fMRI dataset and 4.39% on MS.LotKa-Volterra dataset; 3) single and fixed kernel embedding can not well characterize complex temporal patterns thus leading to poor performance, e.g., decreasing the AUPRC score by 4.89% and 7.90% on VAR dataset and Lorenz-96 dataset, respectively. In contrast, multi-scale adaptive temporal information is essential for recovering LCVs. Additionally, recent studies have also revealed the effectiveness of multi-scale information for time series representation learning [59], [60].

2) *Portability Ability Analysis of CAGKE*: As a key component of MiTCD, CAGKE can be regarded as a general discrete-to-continuous value transformation technique for DVs in MiTS. In this part, portability ability experiments are carried out to analyze the generalization capacity of CAGKE for existing methods based on the Lorenz-96 dataset. Concretely, we transplant the pre-trained CAGKE to other causal discovery methods to obtain LCVs from DVs. Here, we adopt four typical baseline methods (LGC, GVAR, cMLP and cLSTM) for causal discovery from time series processed by CAGKE. Consequently, the causal discovery results assisted by CAGKE are summarized in Table IX. As reported, CAGKE-aided transformation can boost the causal discovery performance to varying degrees compared with directly using the original MiTS. In particular, it brings an AUROC score improvement of 5.77% for GVAR and brings an AUROC score improvement of 5.03% for cLSTM. This is because CAGKE can enrich the information granularity of DVs, thereby facilitating causal interaction modeling in the unified continuous space. Meanwhile, because the operation of CAGKE only requires one forward propagation step, the additional computational effort is relatively small, e.g., about 5.6 seconds per time, which is acceptable for practical applications. The above results demonstrate the great potential of CAGKE in MiTS modeling.

3) *Visualization and Model Interpretation*: To demonstrate the effectiveness of CAGKE, trajectory plots of the original DVs, the recovered LCVs and actual LCVs are presented in Fig. 10, from which we can find that different datasets exhibit different temporal patterns. For example, the DVs of VAR datasets show

densely varied patterns, which are hard to trace and forecast. However, assisted by CAGKE, multi-scale temporal contextual information is incorporated. Thereby, the series becomes more forecastable, promoting the identification of causal relationships. Some cases, such as Lorenz-96 case (ii) and fMRI case (i) show regular periodic patterns. If we predict the DVs directly, the values will *suddenly change* at some time steps, and the *fine-grained continuously changing* patterns cannot be captured, resulting in misleading causal discovery results. Fortunately, empowered by latent continuity recovery, CAGKE gives concern back to the constitutional continuous and fine-grained temporal dynamics of LCVs from DVs. It is beneficial for accurate and reliable estimations of causal structures.

4) *Hyper-Parameter Sensitivity Analysis*: In MiTCD, there are two key hyper-parameters, i.e., the sparsity-induced loss weight λ_3 and the number of multi-scale incentive Gaussian kernels d . In this part, we study their impacts. Fig. 11 provides the model performance under different choices of λ_3 and d on each dataset.

- i) *Impact of λ_3* : λ_3 controls the magnitude of the sparse constraints. On the one hand, too small λ_3 would make the causal variable selection invalid and result in many redundant causations. On the other hand, if λ_3 is too large, the model will be parsimonious thus leading to poor prediction accuracy and low causal discovery ability. Therefore, a proper setting of λ_3 is essential for final results. In the experiments, we finetune λ_3 between the range of $\{5e-3, 1e-2, 2e-2, 5e-2\}$. We find out the best performances are obtained when $\lambda_3 = 1e-2$ or $\lambda_3 = 2e-2$ (Fig. 11 upper), which are in line with the theoretical analysis.
- ii) *Impact of d* : Besides, we present that MiTCD is relatively stable to the number of multi-scale incentive Gaussian kernels d over extensive datasets (Fig. 11 below). Although the performance increases slightly with the increase of d for some cases, we should note that a larger d requires larger memory costs and computational efforts. Practically, in most cases, $d = 10$ is enough to characterize the temporal patterns and guarantee efficiency as well.

5) *Analysis of Discrete Variables With Multiple States*: In reality, DVs can take on multiple discrete states ($N_{DV_s} \geq 2$) that reflect multi-level magnitude. It is noted that more states in a DV imply richer information granularity (with an infinite number of states representing continuous variables). In the main experiment settings, we chose binary DVs (i.e., $N_{DV_s} = 2$) as they are most commonly encountered in the real world and are the most challenging ones due to the minimum information granularity. MiTCD can also process DVs with multiple discrete states effectively. In this part, we investigate its performance under varying N_{DV_s} based on Lorenz-96, VAR, fMRI and MS-LotVol. datasets. To obtain DVs with N_{DV_s} discrete states, we first normalized the original variables to the range of $[0, 1]$ by min-max normalization. Then this interval is divided into N_{DV_s} equal segments as $(0, \frac{1}{N_{DV_s}}, \frac{2}{N_{DV_s}}, \dots, \frac{N_{DV_s}-1}{N_{DV_s}}, 1)$. Each normalized value is subsequently assigned to a discrete state corresponding to the segment in which it falls. The experimental

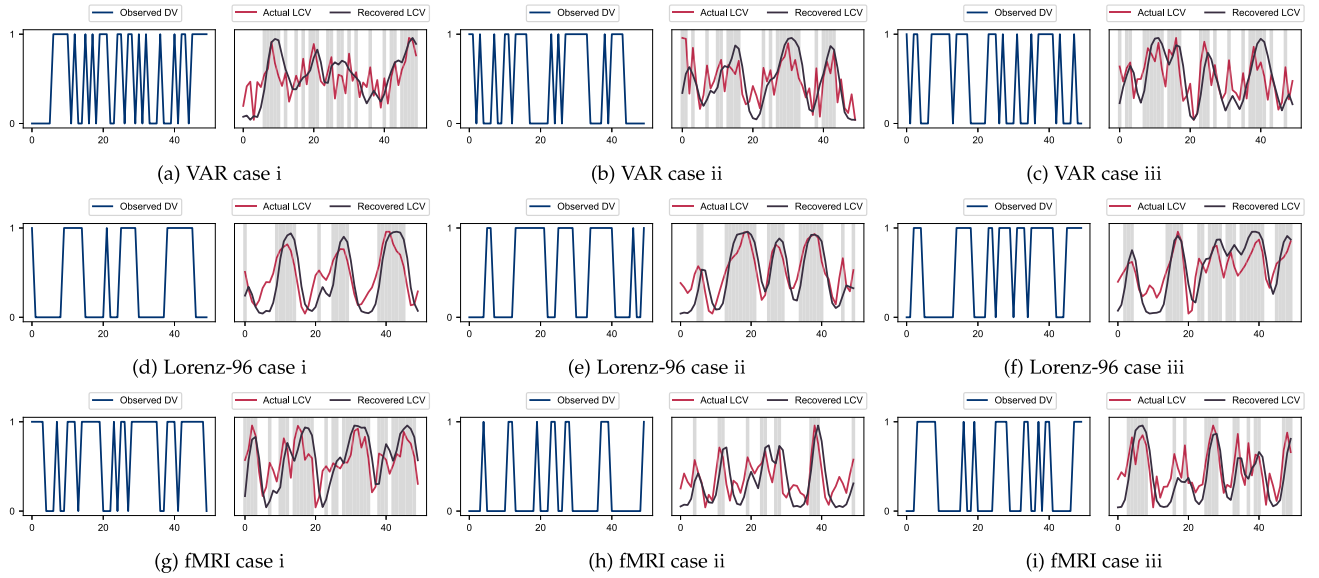


Fig. 10. Visualization of latent continuity recovery on VAR, Lorenz-96, and fMRI datasets. For each sub-figure, the left plot presents the observed DV (with **BLUE** curve), and the right plot presents the recovered LCV (with **BLACK** curve) and the actual LCV (with **RED** curve). The gray rectangular patches denotes the value of the original DV is “1” in the current time stamp.

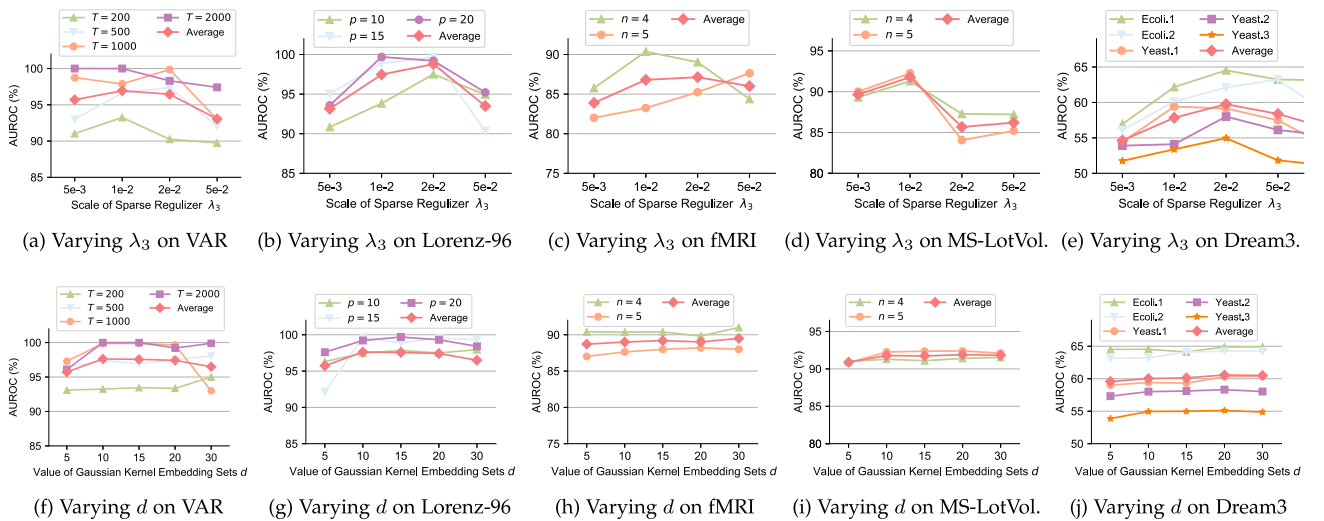


Fig. 11. Causal discovery sensitivity analysis for scale of sparse regularizer λ_3 and value of Gaussian kernel Embedding sets d .

results are summarized in Table X, showing that as the number of discrete states increases, the accuracy of causal discovery increases because more fine-grained information is introduced, which is in line with our expectation.

6) *Analysis of Continuous Variable Scenarios*: Actually, (pure) continuous-valued time series can be regarded as a special case of MiTS. To empirically evaluate this special case, we have conducted the experiments on fMRI dataset and conclude the performance in Fig. 12. To apply MiTCD to the pure continuous-valued time series data causal discovery task, CAGKE and the corresponding pre-training stage are removed since there are no DVs. Specifically, Fig. 12(a) presents the performance

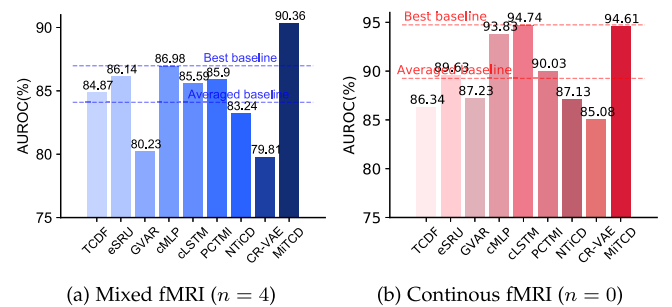


Fig. 12. Performance comparison of mixed and pure continuous variable scenarios.

TABLE X
PERFORMANCE ANALYSIS ON DIFFERENT NUMBER OF DISCRETE
STATES OF DVs

Discrete states	Metric	$N_{DV_s} = 2$		$N_{DV_s} = 4$		$N_{DV_s} = 8$	
		AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC
Dataset	Lorenz-96	96.59	93.78	97.31	94.12	97.66	94.52
	VAR	97.38	93.56	97.68	94.19	98.21	95.31
	fMRI	87.64	78.19	88.42	79.32	88.56	80.31
	MS-LotVol.	91.29	77.69	91.33	79.30	91.89	79.99

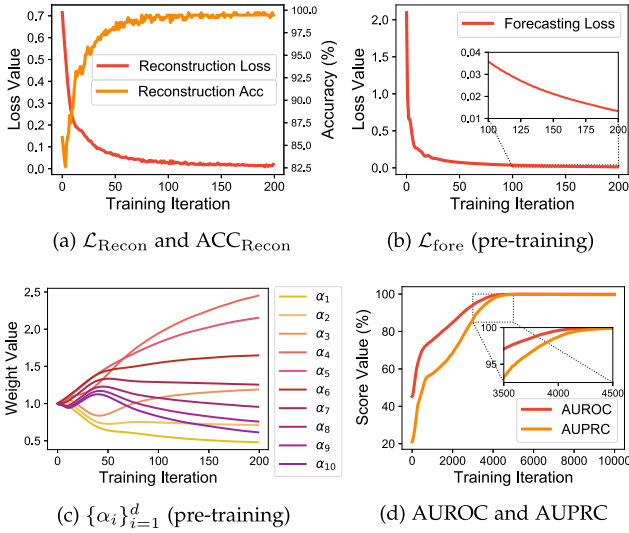


Fig. 13. Model convergence visualization.

comparison on mixed fMRI dataset, from which we can observe that our approach outperforms the second-based baseline method by a large margin (i.e., from 86.98% $\rightarrow$ 90.36%). In contrast, we can see that the superiority of our MiTCD is weakened in the continuous-valued time series scenarios as shown in Fig. 10(b). This is probably because several special designs in our approach for MiTS (e.g., CAGKE and corresponding training strategies, constraints) do not play important roles in scenarios when time series are composed of pure CVs. However, MiTCD still presents competitive results.

7) *Model Convergence Analysis*: To verify the effectiveness of the optimization process, several model convergence results are provided in Fig. 13. Although there are some little fluctuations, the reconstruction loss and forecasting loss are convergent, which verifies that the predictors and discrete variable decoders can be well optimized in the process of pre-training and play their vital roles. Meanwhile, the weight coefficients $\{\alpha_i\}_{i=1}^d$ of CAGKE can also be adaptively learned to characterize different importance of multi-scale incentive Gaussian kernels during training iterations. Besides, as shown in Fig. 13(d), accurate causal structures can be gradually learned during the sparsity-inducing deep causal relation learning stage.

VI. LIMITATIONS AND FUTURE WORK

The key idea of MiTCD is to recover the LCVs for DVs and perform causal discovery in a unified continuous-valued space. Thereby, MiTCD is applicable for *integer discrete variables* whose discrete states represent the physical magnitude of the numerical value. The integer discrete variable is the most common type of discrete variable in time series data, as its discrete states often change over time to reflect the temporal magnitude evolution of the corresponding variable. For example, precipitation levels, health states, and sales demand. As MiTCD mainly focuses on time series causal discovery, a limitation is that it can not be directly applied *categorical discrete variables* whose discrete states represent different and mutually exclusive class labels (e.g., gender, color, and blood type), as they do not have physical magnitude meanings and are often recorded as independent samples instead of time series. In the future, it is of interest to exploit the analysis of generalized mixed variables (including continuous variables and both integer discrete variables and categorical discrete variables) by empowering MiTCD with hybrid modeling strategies, e.g., one-hot embedding and generalized additive models (GAMs) [61].

VII. CONCLUSION

In this paper, we present a generic MiTCD framework for MiTS causal discovery. MiTCD mitigates the research obstacle of Information Asymmetry by recovering the latent continuity from DVs under the guidance of observed CVs. In this way, the gap in information granularity between DVs and CVs can be bridged, and the heterogeneous distribution types can also be unified to facilitate causal discovery in the continuous-valued space. The continuity recovering process is realized via the designed CAGKE technique, which can adaptively aggregate multi-scale temporal contextual information of DVs to re-establish continuous signals by introducing incentive Gaussian kernels. Built upon CAGKE, two inter-dependent training strategies can learn pre-trained optimal model parameters in a self-supervised manner and obtain causal structures for MiTS under sparsity-induced constraints, respectively.

With great generality, MiTCD demonstrates strong performance across multiple benchmark datasets with different settings. In addition, we find that pre-trained CAGKE exhibits superior portability and is compatible with various causal discovery approaches to further promote their application in MiTS modeling tasks.

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